Bachelor Thesis

Designing Hybrid Intelligence Frameworks: Generative AI and Spatial Hypertext for Supporting Human–AI Collaborative Knowledge Systems

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Abstract

In recent years, Generative AI is widely used in knowledge-intensive work like research, writing and analyisis. Large Language Models (LLMs), which are commonly used in Generative AI systems can generate text, summarize documents and support idea development in many domains. But their outputs are probabilistic and they are usually delivered in linear chat interfaces. In general, they do not offer traceable and persistent structures; because of this, generative AI can expand content very efficiently but does not stabilize knowledge into organized representations. On the other hand, spatial hypertext supports emergent and human-centered structures through spatial grouping, clustering and gradual stabilizations. It allows users to externalize changing interpretations without the need of formal schemas. However spatial hypertext does not generate content automatically. This thesis focuses on the structural asymmetry between these two concepts; Generative AI and Spatial Hypertext. The thesis presents a theoretical hybrid intelligence framework that explains how these paradigms can be combined together to support structured human-AI collaboration in knowledge-based systems. This research follows a conceptual/theorethical approach based on systematic literature analysis and comparative study of existing representative systems. The main contribution is the definition of four integration principles: human-centered structural control, emergent-to-formal transition, workspace-mediated AI guidance and iterative co-structuring. These principles are organized into a conceptual architecture that seperates human agent, generative module, spatial/visual workspace and mediation layer. The findings suggest that effective human-AI collaboration develops by synchronizing generative flexibility with persistent and interpretable spatial structures. The theoretical framework in this thesis could offer a clear foundation for the future research and system designs in hybrid intelligence systems for knowledge work.

Contents

Chapter 1

Introduction

1.1 Motivation and Problem Statement

In recent years, Generative artificial intelligence became an important part of knowledge intensive activities in recent years, such as research, writing, planning, and analysis. Large Language Models (LLMs) can generate coherent text, summarize documents and give explanations in many domains [3]. In professional environments, AI copilots are integrated into workflows to assist with drafting, coding, and information processing [5]. These systems reduce effort in early-stage work and accelerate idea generation.

However, generative AI has structural limitations in knowledge work. LLM outputs are probabilistic and typically presented in linear, chat-based interfaces [3]. Their internal representations are not directly inspectable, and they do not provide persistent conceptual structures that users can manipulate [7]. As a result, generative AI produces fluent content, but it does not automatically stabilize knowledge into organized, durable representations. Outputs appear as sequences of text rather than as visible, evolving structures.

Complex knowledge work requires more than text generation. Research on spatial hypertext shows that understanding often develops through grouping, clustering, and rearranging information in space [14, 15]. Structure is not always predefined , instead it develops gradually through interaction. Users place related items close to each other, separate unrelated elements, and slowly adjust clusters over time. This process helps people understand information in situations where knowledge is incomplete or unclear.

Recent systems further highlight a mismatch between linear chat interfaces and nonlinear reasoning processes. For example, Sensecape demonstrates that sequential conversation makes it difficult to maintain overview, revisit earlier ideas, and work across multiple abstraction levels [21]. Effective knowledge work requires persistent and visible structure, not only conversational exchange.

Despite these complementary insights, research on generative AI and research on spatial hypertext have largely developed separately. Generative AI research focuses on model capabilities and prompt interaction [3, 7]. Spatial hypertext research focuses on human-centered structure and emergent organization [13, 14]. Some systems already started to combine these ideas, like SPORE [19], Sensecape [21], and space-steered summarization paper [22]. But in general, these integrations are domain specific or partially integrated. They do not provide a general theoretical explanation of how AI and spatial hypertext can be systematically combined as one system. This reveals a structural gap where generative AI offers generative flexibility without a stable structure while spatial hypertext provides stable structuring without generative capabilities. What is missing is a complete meaningful theoretical framework which explains how these two strengths can be integrated in a principled and human-centered way.

1.2 Research Aim and Questions

The aim of this thesis is to develop a theoretical hybrid intelligence framework that explains how generative AI and spatial hypertext can jointly support human–AI collaboration in knowledge-based systems.

The thesis does not propose a new algorithm or a production-ready system. Instead, it focuses on conceptual integration. It seeks to clarify how generative flexibility and spatial structural stabilization can be combined without losing human interpretability or control.

The thesis is guided by the following research questions:

- **RQ1:** How can generative AI and spatial hypertext be structurally combined to support human–AI collaboration in knowledge-based systems?
- **RQ2:** What roles do spatial organization, emergent structure, and informal representation play in hybrid intelligence frameworks?
- **RQ3:** How can spatial hypertext function as a shared cognitive workspace that mediates interaction between humans and generative AI?

These questions structure the argument of the thesis. The first question focuses on integration logic. The second examines the structural and cognitive contribution of spatial hypertext. The third analyzes the mediating role of the workspace in hybrid systems.

1.3 Methodological Approach

This thesis follows a conceptual methodology based on structured analysis of literature in four related domains: generative AI and large language models, spatial hypertext and visual sensemaking, hybrid intelligence theory, and AI-supported knowledge work systems. Instead of conducting new empirical experiments, the thesis synthesizes findings from prior research and compares existing systems. Systems such as SPORE, Sensecape, [22], and EDGE are analyzed through a shared analytical perspective derived from the structural conditions discussed in earlier chapters. From this analysis, four core principles of hybrid intelligence are formulated: human-centered structural control, emergent-to-formal transition, workspace-mediated AI guidance, and iterative co-structuring. These principles are then organized into a conceptual architecture that explains how generative modules, spatial workspaces, mediation layers, and human agency interact. The main contribution of the thesis is therefore theoretical and structural, clarifying integration conditions and proposing a framework that can guide future system design and analysis.

1.4 Thesis Contribution

This thesis presents three main contributions to the study of hybrid intelligence in knowledge-based systems.

Primarily, it gives a clear and structured analysis of the connection between generative AI and structure without content (spatial hypertext). Although both areas have been studied a lot, they are gnerally explored separately. This thesis looks at them altogether and points out a structural asymmetries: generative AI allows fast union but does not provide consistant and inspectable structure, while spatial hypertext provides visible and user-controlled structure but does not include generative capabilities. By explaining this asymmetry clearly, the thesis defines the main integration problem. Furthermore, the thesis derives four design principles for hybrid intelligence systems. These principles are grounded in literature and comparative analysis of existing systems. They are:

- Human-centered structural control

- Emergent-to-formal transition
- Workspace-mediated AI guidance
- Iterative co-structuring

These principles transform conceptual analysis about "AI collaboration" into concrete structural conditions. They define how Generative AI and spatial workspaces should interact in order to support stable and understandable knowledge development.

Lastly, the thesis introduces a conceptual architecture that puts these principles into practice. The architecture separates four main components: the human agent, the generative AI module, the spatial hypertext workspace, and a mediation layer that converts spatial configuration into computational context. This layered model makes it understandable how interaction should be structured and how tasks are shared between humans and the machine.

It is important to note that this thesis does not present a completed software system or a fully validated experimental model. Its main contribution is theoretical and structural. It provides a basic framework that can help guide future system design, evaluation, and research in hybrid intelligence for knowledge work.

1.5 Structure of the Thesis

This thesis includes seven chapters. Each chapter builds on top of each other in a systematical manner and step by step develops the proposed hybrid intelligence framework.

Chapter 2 examines generative AI in knowledge-based systems, looking at the capabilities of Large Language Models, such as drafting, summarization, and synthesis. At the same time, it points out structural limitations, including probabilistic variability, the lack of inspectable internal representations, and reliance on sequential chat interfaces. This chapter explains why generative AI by itself is not enough for structured and continuously developing knowledge work.

Chapter 3 introduces spatial hypertext as a complementary paradigm. It explains main concepts such as implicit structure, spatial grouping, emergent organization, and incremental formalization. The chapter also discusses systems such as VIKI and Mother to show how spatial workspaces help human-centered sensemaking. This chapter explains how structure can develop step by step through interaction.

Chapter 4 connects the two domains. It compares generative AI and spatial hypertext and points out a structural asymmetry between generative breadth and structural persistence. Based on this comparison, the chapter describes the integration conditions that any hybrid intelligence system needs to achieve. This chapter defines the main conceptual problem area of the thesis.

Chapter 5 describes the proposed hybrid intelligence framework. It defines four design principles and explains a conceptual architecture consisting of a human agent, a generative module, a spatial workspace, and a mediation layer. The chapter explains how these components work together through iterative co-structuring and step-by-step stabilization.

Chapter 6 applies the framework as an analytical lens. It examines representative systems such as SPORE, Sensecape, [22], EDGE, and an exploratory prototype. Rather than repeating literature review, this chapter evaluates how each system realizes or omits the proposed principles. This comparative analysis demonstrates the relevance and necessity of the framework.

Chapter 7 combines the findings of this thesis answering the research questions, discussing theoretical and practical implications, identifying limitations, and providing directions for future research.

This way the thesis moves from studying existing paradigms to extracting integration principles and finally to a structured synthesis.

1.6 Positioning and Broader Relevance

The rapid adoption of generative AI in professional and academic environments makes it increasingly important to understand how such systems should be designed for long-term knowledge development rather than short-term content production. While current tools often emphasize speed and convenience, knowledge-based systems require persistence, interpretability, and structural stability.

This thesis presents hybrid intelligence as a clear and structured alternative to only using conversational AI interaction. By combining generative flexibility with spatial stabilization, the proposed framework adds to current discussions about human–AI collaboration, cognitive augmentation, and responsible system design. It shows that effective AI support in knowledge work depends not only on model capability, but also on how interaction is organized and how information is represented.

The framework described in this thesis may support future research on AI-supported workspaces, intelligent visual interfaces, and knowledge management systems.

Chapter 2

Conversational and Generative AI

2.1 Conversational AI

2.1.1 Evolution of Conversational AI

Conversational AI, as a subfield of artificial intelligence, is usually described as systems that interact with users through text-based or speech-based natural language conversation [1]. In the literature, Conversational AI Agents are sometimes also referred to as chatbots [1]. The main goal of these systems is to simulate or automate human-computer interaction by allowing users to communicate with them using everyday language/natural language rather than specialized commands or complex interfaces [1]. This capability made Conversational AIs very popular as an interface for knowledge-based systems, where users need to access, query, or manipulate information flexibly and naturally [4]. The usage of Conversational Interfaces have been mostly useful in domains where it relies on heavy language-based interactions including education, healthcare, customer support and information services [4],[5]. Advancements in machine learning, deep learning and increased computational power in recent years also greatly influenced the rapid growth of Conversational AIs [1].

Early research (1950s) on Conversational AI was mainly influenced by the idea that the artificial intelligence could be evaluated by its conversational capabilities and language-based interactions. [4] This idea, named 'The Turing Test' proposed that the machine could be considered intelligent if its conversational behaviour is not distinguishable from the humans. [4] Thus, the earlier research on chatbots mainly focused on the imitation of human behaviour rather than reliable understanding or task support. The first chatbot named ‘ELIZA’, developed in 1966, relied on pattern matching and template-based response selection scheme to simulate dialogues. [4] Its capabilities in communicating was very limited, although it influenced the development of chatbots. [4] Even with its simplicity, ELIZA showed that the basic language rules could create the impression of understanding even when conversations were limited. [4] Later developed chatbot system named ’PARRY’ (1972) expanded on this approach by simulating a specific personality and assumptions, showing better conversational behaviour in different evaluation environments. [4]

Following these systems, many chatbots were developed using rule-based approaches, where predefined patterns were manually assigned to the predefined responses. [4]

Using programming languages like AIML (Artificial Intelligence Markup Language) allowed developers to create large rule sets in order to allow chatbots to handle predictable conversational scenarious in more effective ways. AIML helps set up conversation patterns and responses, so chatbots can understand and reply to users in a natural, human-like way. [4] In 1995, ALICE was one of the chatbots developed by using this language and ALICE’s Knowledge Base consisted of ’41,000’ templates and rules, meanwhile ELIZA had only ’200’ rules and keywords [4].

Although rule-based systems performed good in limited and predictable domains, they usually struggled to handle linguistic variation, uncertainty, and unexpected user input [4]. Additionally, the manual effort that was needed to maintain and extend these systems limited their scalability and adaptability [4]. Later on, the Conversational AIs shifted from imitation based systems to more task oriented and information centered interactions. [4] Rather than only focusing on having the realistic conversation, later systems/chatbots integrated conversational interfaces with databases, APIs (Application Programming Interfaces), and backend services to support information retrieval and knowledge management [4]. This change in focus transitioned Conversational AIs from conversation as an end goal to conversation as a means of accessing information and applying knowledge.

Despite all these advances, evaluating Conversation AI has been a major challenge according to the literature [2]. Conversation quality is highly dependent on the user expectations, system goals and usage context, making universal evaluation metrics impractical [2]. A system that performs well according to technical metrics may still fail to meet user needs in real world interaction scenarios [2]. These difficulties in evaluation shows that conversational capabilities are not sufficient enough to guarantee effective support in knowledge based work, and that further developments in conversational interfaces are required [2].

2.1.2 Core Architectures of Conversational Systems

A commonly used approach to describe conversational AI in the literature is through a modular pipeline architecture [1]. This architecture organizes the system into three separate components and each component is responsible for specific part of the conversational process:

- Natural Language Understanding (NLU),
- Dialogue Management (DM),
- Natural Language Generation (NLG) [1]

A modular structure allows developers to easily separate responsibilities within the system. It also allows to develop and improve specific components individually whenever needed [1].

The Natural Language Understanding (NLU) component is responsible for understanding the input that is in the form of unstructured text. [1] Its main task is to transform the unstructured text into a structured format which the system can process. [1] Even with common human mistakes in the conversation like misspellings, typos (typographical error). or mispronunciations, NLU is capable of extracting the meaning in them. [1] Two main task used in NLU commonly include intent classification and named entity recognition. [1] The goal of Intent Classification is to identify the reason or the objective behind a user's statement, such as requesting information or performing an action. [1] Named entity recognition focuses on detecting and classifying relevant elements in the input like; names of people, locations, organizations, or domain specific concepts [1].

Once the NLU interpretes the user input, the response that will be given by the system is determined through the Dialouge Management (DM) component. [1] The Dialogue Management component considers the current message, previous messages in the conversation, and the system's objectives before deciding what to do next. [1] Because DM keeps track of the dialogue state, the system can manage multi-turn interactions instead of responding to each message separately. [1] In task-oriented systems, DM leads users through structured conversational steps so that specific goals such as booking or retrieving information can be completed. [1] According to the literature, dialogue management can be defined as a sequential decision-making process that can be implemented using approaches like; finite-state machines, reinforcement learning, or probabilistic models. [1]

After the Dialouge Management component, the Natural Language Generation (NLG) component creates the system's final output/response in a human-readable language. [1] Opposite to NLU, which converts the user input into structured data, NLG converts structured system information back into meaningful natural language responses. [1] Earlier NLG systems used predefined rule-based templates to generate responses, which made outputs more consistent and predictable but also made it limited in variation and flexibility. [1] More recent NLG systems generally use neural-network based generation models which are able to produce more natural, fluent, and diverse language. [1] While they provide more natural, fluent and diverse language, neural-network based models can make it harder for developers to fully control and predict the outputs of the system, making it less flexible in this manner. [1]

Addition to the main conversational pipeline (NLU, DM, and NLG), many conversational AIs also integrate backend components like; databases, retrieval systems, and knowledge bases to further improve their functionalities. [4] Backend systems allow conversational AIs to access, store, and update structured information outside the dialogue state [4]. This architectural integration becomes especially important in systems designed to manage structured knowledge, where conversational input must be transformed into consistent internal representations. [1] [11]

2.1.3 Response Generation Paradigms

An important step in conversational AIs is how the system generates a response after it understands the user input and it selected the next action (NLU, and DL). In literature [4], few generation paradigms are described. These include; rule-based, retrieval-based, generative and additionally hybrid approaches [4]. These approaches are different in ways of how flexible, how controllable and in which application domains they can be effectively used. Rule-based generation is based on predefined templates and patterns [4]. In this approach user inputs are directly linked to the predefined responses meaning that the system can only answer user queries based on the rules that were manually set. Because of this structure, rule-based systems are usually faster in speed, and are predictable and more controllable [4]. On the other hand, they are only limited to what they have been programmed to and in general cannot deal with unexpected inputs or complex language variations [4]. Retrieval-based models use a neural network to assign scores and select the most likely response from a set of responses based on the similarity between the user input and stored responses [4]. The systems selects the most appropriate response to the user input from existing repository [4]. With this approach, the responses can be contextually more relevant in specific domains, especially when high quality response databases are available in the approach [4]. Still, retrieval-based systems are also limited with the responses that are available in the database and cannot generate new and more creative responses [4]. Generative based systems on the other hand produces new responses by modeling language statistically instead of selecting from predefined and fixed options [3], [4]. Neural generative models are trained on large datasets and learn patterns in natural language that allows them to generate flexible and more diverse responses [3]. While generative-based systems are useful in open-domain conversations, they are usually not ideal for domain-specific conversations because they may produce incorrect or inconsistent responses [4]. To solve the weakenesses of these systems, hybrid approaches are used in many conversation AI systems [4]. Hybrid systems combine rule-based or retrieval-based methods with generative models [4]. This allows the system to balance between structured situations while using generative techniques for more open ended conversations [4]. For example, a system may rely on predefined reules or retrieval methods for frequently asked questions (FAQ) and can also switch to generative responses when there are no suitable answer for the question asked [4]. The need for hybrid methods highlight the need to balance the naturalness of conversations with dependable behaviours in knowledge-based systems [1], [4].

2.2 Generative AI and LLMs in Knowledge Work

2.2.1 Generative AI Foundations

Generative artificial intelligence (GenAI) is a type of AI system that creates new content instead of just analyzing or classifying existing data[3]. This content can be text, images, audio, video or even source code depending on the application domain [3]. Unlike the traditional AI systems that mainly focus on prediction, decision making or simply pattern matching, generative AI produces new ouptuts resembling human created data [3]. Because of this capability, generative AI shows and important change in how AI systems interact with user and information.

To better understand this shift, it is useful to briefly contrast generative AI with earlier symbolic and rule-based approaches, such as expert systems [6]. Expert systems relied on manually encoded knowledge bases and predefined inference rules to imitate human expert decision-making [6]. While they worked good in limited and specific domains, they struggled with scalability, flexibility, and updating knowledge as problems became more complex [6]. On the other hand, generative AIs learn patterns directly from larger datasets and they do not depend on pre-defined rules which allows them to handle a broader range of inputs [6].

In literature, there are various types/families of generative AI forming the base of generative AI systems [3]. Variational Autoencoders (VAEs) are probabilistic models that learn to compress input data (like images or text) into a compact latent representation and then reconstructs it to generate new samples [3]. Generative Adversarial Networks (GANs) consists of two components; generator and discriminator that are trained together [3]. The generator component tries to produce realistic outputs while the discriminator tries to detect if the output is real or AI generated [3]. Autoregressive models generate data step by step by predicting the next element based on the previous elements, which makes them a better fit for language modeling tasks while it represents a conversational behaviour [3]. Diffusion models create outputs by slowly removing noise that was added during the training, achieving strong generation results in various domains [3].

Even though generative AIs have strong capabilities in many application domains, they still have limitations. Generative AI systems does not posses the real semantic understanding or human like reasoning capabilities [6]. Their behaviour is driven by statistical pattern recognition learned from the training data [6]. As a result generative AI systems can come up with outputs that appear meaningful and correct without understanding the underlying concepts [6]. This is limitation is important in knowledge-based systems while these systems require accuracy, correctness, reasoning and interpretability.

2.2.2 Large Language Models as Knowledge Assistants

Large Language Models (LLMs) are described as an important type of generative AI, especially for text based and knowledge related tasks [3]. LLMs are trained on huge datasets of text using self-supervised learning that allows them to learn statistical patterns in natural language [3]. Many of the recent LLMs are built on transformer architectures that helps to understand relationships between words even if they are far away in the sentence or document [3]. Since they are trained on large amounts of textual data, they can perform many knowledge related tasks. These tasks include text generation, summarisation translation, question answering and content explanation [3]. LLms are usually described as general purpose assistants that helps users with writing documents, combining information or exploring new topics [3]. For this reason LLMs are useful for knowledge-based systems that require natural language reasoning nad flexible interaction capabilities [3].

However LLMs still showcase several limitation when used in knowledge work. One of the main issues is that LLMs can generate responses that sound convincing as if it is true, but can actually be misleading or false [3]. This problem is commonly known as hallucination [3]. Additionaly LLM outpus are probabilistic meaning that the same prompt can produce different responses each time [3]. This affects reliability and consistency. Finally LLMs do not have background understanding and they do not automatically check their answers for proof using external knowledge sources unless they are connected with such systems [3].

In recent years LLMs have been integrated into everyday digital tools leading to the development of AI assistants that operate within the existing workflows [5]. These assistants are often called AI copiltos and they support users in tasks like; writing, programming, and analyzing information [5]. Kanmigo is one of example of a generative AI assistant which is used in the field of education, where the system focuses on guiding the user instead of directly giving the answers [10]. Research shows that the effectiveness of such systems depends on the factors such as how appropriate the tast is, the level of expertise and the design of the interaction [10].

2.2.3 AI Copilots and Generative Agents

Recent research shows that the way AI systems are viewed in knowledge-based environments has changed. Instead of just being automation tools that perform tasks AI systems are started to be described as collaborative assistants that works with humans [5]. This change in human-ai interaction shows that AI is no longer meant to be a replacement for human work but rather support and improve human cognitive activities [5]. Within this context, AI copilots are described as AI systems that share cognitive tasks with users instead of simply executing commands [5]. Tasks can include writing text, suggesting actions, guiding user through complex processes or reviewing content. This is a representation of augmentation oriented interaction model where AI assists but does not fully replace human judgment[5], [19]. One of the main feature of AI copilots is that they are generally integrated into existing systems and workflows. The literature [5] explains that these ststems are usually embedded directly into existing tools such as document editors, programming environments, enterprise platforms, etc. [5] While they operate inside of these environments, they can also access contextual information such as; previous documents, user activity or taks history which then allows them to provide more relevant and useful assistance on demand [5]. In AI copiltos, the effectiveness of the system usually relies on factors such as; how clearly user prompts the system, how specific the domain is or how experienced the user is with the system or environment [5].

Beyond the general purpose systems, the literature [6] introduces a concept called Generative Artificial Experts (GAEs) which is a specialised form of generative AI for knowledge work [6]. GAEs are domain-specific assistants that supports users with complex tasks within a specific domain while still being under the human supervision [6]. Unlike the general purpose LLMs which aim to cover many/diverse topics, GAEs usually focus on deeper knowledge inside a specific professional domain/field [6]. However the literature [6] says that GAEs are still mostly conceptual and that they are not yet fully yet developed or implemented systems [6]. In the field of knowledge management, intelligent assistants started to emerge as essential tools that support knowledge processes like; knowledge creation, retrieval, sharing and etc. [7]. These systems help users to organize, filter and present information especially in environments where information overload is common [7]. These assistants are mostly effective when they are designed to support human processess instead of fully replacing the human decision making [7].

2.3 Human–AI Collaboration in Knowledge-Based Systems

2.3.1 From Automation to Collaboration

Earlier AI systems were mainly used to automate tasks in organizations and industry and were repetetive and routine [5]. These systems were predefined and operated in controlled environments. Main goals of these systems were to replace human labor in specific processes rather supporting human decision making processess [5]. With the development of generative AI way AI systems are understood in knowledge related work changed [5]. Instead of being seen as only automation tools, these systems are nowadays are viewed as systems that can support and augment human work [19]. The focus has changed from replacing humans to collaborating with them [19]. The literature describes this change as "AI as a tool"to "AI as collaborator" in knowledge based environments [5].

From this perspective, human-AI interaction is no longer viewed as a simple sequence where humans provide an input and machines produce an output for that input [8]. Instead both human and AI systems contribute on completing a task where both their actions influence each other [8]. Their activities are interdependent and a good performance depends on how well they coordinate [8]. This idea is a reflected in the concept of hybrid intellingence systems which combines human intelligence and AI to achieve outcomes that they would not achieve alone [8]. Another aspect of this perspective is the focus on collective performance and mutual learning [8] Rather than evaluating AI systems by technical accuracy it is suggested to evaluate how human and AI systems interact, adapt and learn together [8] In knowledge work, AI systems should function as partners that support human expertise instead of replacing it [7]. Knowledge work involves interpretation, judgement and contextual understanding which are mainly human capabilities [7]. At the same time AI systems can assist by analyzing large datasets, identifying patters and providing computational support which can be diffuclt for humans to deal with alone [7]. Therefore effective human-AI collaboration needs placing machine capabilities with human strength and responsibilities [7]. This argument outlines a core principle of collaborative and hybrid intelligence systems which sets the foundation of the following sections and the thesis'focus.

2.3.2 Division of Cognitive Labor Between Humans and AI

How the cognitive labor is divided between human and AI systems is an important factor of human-AI collaboration in knowledge-based systems. The literature distinguishes between specialized intelligence and general intelligence [7]. Specialized intelligence is usually associated with AI systems and general intelligence is mainly human centered [7]. This disctinction explains how human and AI can complement each other instead of working for the same tasks. AI systems are strong in task specific operations like pattern recognition, classification and processing large amounts of data [7]. AI analyzes information faster and detect patterns that humans may not realize or see [7]. But these strengths are usually limited to specific tasks [7]. Humans on the other hand are better at general reasoning, ethical judgement and contextual interpretations [8]. Knowledge work usually requires understanding implicit meanings, considering situational context and creativly framing problems, which are main human capabilities [7]. Because of these differences, collaborative systems should be designed to balance and use strengths of both human and AI [7]. In such systems AI and human contribute where they are most effective [7]. In this context the concept of bounded autonomy comes in place [6]. Bounded systems are AI systems that can perform independently in certain tasks while being under human supervision [6]. The idea is to balance control and efficiency by letting AI operate within certain limits instead of fully giving it autonomy [6]. This is an important point when it comes to knowledge-based works where false outcome can lead to serious consequences [6]. The final responsibility should always remain with humas especially in important and complex decision making applications [6]. Therefore human oversight is a key desing principle in collaborative AI systems. It makes sure that AI generated outputs are reviewed, changed if needed and validated by users before finally applying it [6]. Another aspect of cognitive labor division is learning and adapting over time. In collaborative systems, both humans and AI can learn throughout the interaction and feedbacks [8]. AI systems can adapt their behaviour based on what user corrects overtime, new data or changing task requirements [8]. At the same time humans refine their understanding while interacting with AI outcomes and explanations [8].

2.3.3 Specialized Assistants and Collaborative Agents

Building onto the previous discussions of human-AI collaboration and division of cognitive labor, knowledge-based systems generally require more specialized AI assistants rather than general-purpose models. In more complex environments AI systems need to get along with specific domains and tasks in order to provide reliable and controlled support. As already explain in earlier sections, generative artificial experts and AI copilots are two examples of such systems that align with these needs [5, 6]. GAEs mainly focus on giving expert level help/assistance within a specified domain and they operate under human supervision [6]. But GAEs are still mostly conceptual and they do not provide much practical implications yet [6]. On the other hand, AI copilots are already implemented in many practical systems and professional tools like document editors, software development environments and more [5]. Integration of these systems provide context aware support. At the same time, increased specialization requires careful design to make sure reliability [5]. Specialization also appears in the form of guided or scaffolding interaction, with this approach AI systems do not provide direct answers but support user with hints, guiding user questions and provides structured assistance [10]. This is more important in learning contexts or knowledge building applications where understanding and reasoning is more important than just toreceive a final answer [10]. Overall specialization shows that effective human-AI collaboration is not only about powerful models but also about clearly defined scope, accurate integration and balancing human control [5], [6], [10].

2.4 Limitations of Generative AI in Knowledge-Based Work

2.4.1 Reliability and Uncertainty

Generative AI systems can produce coherent language, but they may also generate incorrect information [3]. This problem we have already mentioned in earlier sections called "hallucination" is considered one of the main problems and challenges of generative AIs [3]. The output may appear confident and fluent even when the information is wrong or unsupported [3]. In these systems, hallucinations can happen because generative models are trained to predicts statistically accurate words sequencences instead of verified facts [3]. Unless external knowledge sources are integrated into such systems, these models do not automatically check if the outputs are true [3]. This is problematic in knowledge-based work where the false information can affect the decision making processes and understanding. Another issue which was mentioned earlier is that these systems generate text by sampling from probabilistic distributions and the exact same prompt can produce different outputs in different runs which affects consistency and reproducibility, which are important details in professional and more serious contexts [3]. Output variability can also depend on the model design, sampling methods and promping of these systems [9]. The concept of semantic uncertainy principle has been proposed to describe this limitations as an analogy to quantum mechanics suggesting that there is a trade off between internal semantic uncertainty in the model and the variabiliy of its outputs [9]. Simply put, uncertainty in generative AI cannot be comeepletly removed even with careful prompting or parameter tuning [9]. That is why fully predictable behaviour is difficult to achieve in complex knowledge environments [9]. Finally, generative AIs do not have real semantic understanding and their outputs are based on statistical pattern recognition which is learned from the training data rather than true understanding [6] This means they can produce responses like experts without really understanding the meaning of the content [6]. Overall, reliability and uncertainty are main limitations when it comes to usage of generative AIs in knowledge-based systems.

2.4.2 Evaluation Challenges

Few aspects of evaluating conversational AIs have been discussed earlier in this chapter. Extending on it, evaluating generative AI systems is difficult beacuse their outputs are open ended [2]. Unlike the traditional conversational systems whre answers can be checked/compared to clear correct results, generative AI does not always produce a single response [2]. Because of this, evaluation cannot really rely on accuracy metric [2]. Performances of generative AI also depends on context, user expectations and the purpose of use [2]. When a systems performs good in one situation it may not perform good in another [2]. For example a creative response can be useful in exploration tasks but not accurate in decision making context [2]. Because of this, it is difficult to define a single universal standard of "good performance" in knowledge-based systems when it comes to evaluating generative AIs [2]. Multi-perspective evaluation approaches are proposed to deal with this complexity [2]. This approaches include user experience, information accuracy, language quality and perceive intellugence [2]. But still a system can perform good in one aspect while failing in another [2]. For example it can produce a seemingly good language and provide inaccurate information, making evaluation more complicated [2]. Khanmigo shows these challenges in educational applications showing that effectives depends not only on the output quality but also on if the system behaviour supports well in learning goals [10]. Systems that provide straightforwards answer may redice learning outcomes when compared to systems that guides users through reasoning using scaffolding techniques [10]. That is why the evaluation should consider the impact of generative AIs on user understaind, engagement and cognitive development instead of only focusing on the correctness/accuracy [10]. Earlier chatbots mentioned in the first section (2.1) also shows important limitations when it comes to evaluation [4]. Many of the earlier evaluations focused mainly on how the conversation feels natural or humanly, especially in Turing test settings [4]. Those systems sometimes approved conversational fluency instead of accuracy of the provided responses resulting in systems which appear intelligent meanwhile providing limited practical values [4]. Overall, evaluating generative AI in knowledge-based environments require taking into account the context, the purpose of the system and user impact and that traditional metrics for evaluation are not enough.

2.4.3 Context, Grounding, and Explainability

Another limitation of generative AI in knowledge-based systems is that they lack explicit grounding in structured knowledge. Generative models mainly rely on the knowledge that is implicitly stored in their parameters rather than on clear knowledge structures [7]. This allows them to create fluent and contextually well language, it makes it difficult to explain or reason on how it came up with the specific output, as a result the user may get a convincing answers from the AI without understanding how the AI came up with it [7]. These system does not explicitly refer to structured knowledge sources therefore users often cannot fully trace the origin or the reasoning behind generated outputs making validation and verification difficult [7]. This can be a problem in knowledge-based environments since trust and accountability in such environments is important, resulting on user having convincing answers from AI without understanding the justification [7]. A distinction between 'know-what' and 'know-why' clarifies this problem [7]. Know-what involves producing factual information, meanwhile know-why involves explaning the reason and providing justification [7]. Generative AIs are usually better at providing know-what outpus and not so efficient in providing know-why explanations in their responses [7]. While knowledge-based tasks require traceability and reasoning rather than answers that seem correct, this aspect of generative AIs become a problem [7].

Knowledge-based systems often depend on structured elements like entities, relationships and conceptual linking [11]. Relying on natural language text makes it difficult to build consistent/reusable knowledge structures [11]. Because of this reason specialized information extraction components are suggested to support for creation of structured knowledge representations with the generative models [11].

Another issue is the sensitivity of generative AIs when it comes to propmting [9]. Tiny changes within the prompts can affect the output quality [9]. Too much or poorly organized context can sometimes reduce performance in more complex environments (a phenomenon described as context overload) [9]. Overall, context handling, grounding and explainability still remains as open research challenges for generative AI and these limitations shows that complementary solutions are needed to support structure, transparency and human centered control.

2.4.4 Risks, Overreliance, and Governance Concerns

When generative AIs are used in everyday knowledge work, the risks are not only about technicality. They also affect people/users, their thinking and organizations. These risks become serious when AI systems are integrated into workflows and decision making processes as it can affect how people think, work and take responsibilities. One important risk is overrelience on artificial intelligence [5]. When users often rely on generative AI to do their cognitive tasks for them, they may become less engaged in analyzing information, questining or considering alternatives [5]. After some time this can lead to deskilling, where users start to lose their capabilites in tasks they regularly let AI do over the time [5]. Similar to overrelience, cognitive complacency is another problem [7]. Cognitive complacency happens when users accept the AI generated results without carefully verifying or evaluating them [7].

Because generative AIs often generate confident and fluent responses, users can believe the responses are correct even if they are not [7]. Addition to behavioural risks, generative AIs also carry security and misuse risks (ethical) [4]. They can be used to create misleading or harmful content with purpose or accidentally [4]. Other risks also include prompt injection, data leaks and accidentally sharing sensitive information [4]. For this reasons, strict rules and responsibilities should be applied when organizations use generative AIs [4]. Human supervision and transparency are important to prevent ethical issues when using these systems within organizations [4]. Generative AI therefore should not treated as a system thatn can work completely on its own, it should support humans and not replace them.

2.5 Interim Synthesis: Requirements for Human–AI Knowledge Collaboration

This chapter examined conversational AI architectures [1], generative AI foundations [3], large language models [3], AI copilots [5] and collaborative agents [5]. Alltogether, this chapter shows that generative AI systems have strong capabilities for supporting knowledge-based work, but they also have several limitations which prevents them from working as independent knowledge authorities. Based on these analysis several key requirements for effective human-AI knowledge collaboration can be deriven. On the useful aspects, generative AI systems, especially LLMs, allow natural language interaction, flexible text generation and quick processing of large sizes data/information [3]. When integrated into workflows as AI copilots these systems can assist users with their writing, programming, analysis and many other knowedge intensive tasks [5]. Their ability to synthesize information and produce coherent responses make them useful tools for supporting knowledge processes like drafting, summarizing, and context explanation [3], [7]. However it is also seen in this chapter that these systems have significant limitations as well. They can generate factually incorrect or misleading outputs (hallucination) [3]. Their probabilistic nature can cause output variability meaning that single prompt can result in different repsonses [9]. Also these systems lack genuine semantic understanding and they rely on statistical pattern recognition rather than proved reasoning [6]. Moreover limited transparency, weak reasoning, and sensitivity to prompt formulation are also included as challenges [7], [9]. While correctness, accountability and traceability are very important in knowledge-based systems, these limitations cannot be ignored. Addition to the technical limitations, human related risks should also be considered. Overrelience in generative AI may reduce cognitive engagement and reflective thinking [5]. Users may accept AI responses without enough verification when the responses appear fluent and confident [7]. As a result, these risks shows that generative AI should not replace humans but instead collaborate with them in supportive role with clear boundaries.

All together, this synthesis leads to several design requirements for human-AI knowledge collaboration:

- Human-in-the-loop control: AI systems need to operate under human supervision, with final responsibility remaining with the user [6][7].
- Transparency and inspectability: Users should be able to examine, question, and validate AI-generated outputs [7].
- Support for structured knowledge representations: Knowledge-based work requires more than fluent text; it often depends on explicit entities, relations, and conceptual organization [11].
- Explicit handling of uncertainty: Variability and probabilistic behavior should be acknowledged and managed and not ignored [9].

Complementary division of cognitive labor: AI systems should focus on computational and generative tasks, while humans retain responsibility for interpretation, contextualization, and judgment [8][6].

These requirements shows that generative AI alone is not enough to fully support collaborative knowledge work, even though such systems are powerful language generators, they

require complementary mechanisms that can provide structure, context management and interpretable representation of knowledge. This motivates the exploration of approaches that goes beyon pure generative interaction. Particularly knowledge-based environments benefit from systems that can support explicit organization of information, visual structuring and interative sensemaking processes. In the next chapter spatial hypertext will be introduced as a complementary framework that can adress structural and organizational challenges that are not fully resolved by generative AIs alone.

Chapter 3

Spatial Hypertext

3.1 Hypertext Evolution and Definition

3.1.1 Spatial Hypertext as an Alternative to Explicit Links

Early Hypertext was first imagined as interconnected reference materials and literature [13]. Later on it turned into real systems with link-based navigation between and within documents [13]. As the hypertext networks grew larger in size and complexity, researchers saw that users could become confused and could get lost (the classic "lost in hyperspace" problem) navigating these large networks, which made navigation support a big concern [13]. This problem started the development of navigation aids and structural overviews.

One of the earlier developments to overcome this issue was with map-based hypertext systems [13]. In systems like NoteCards, maps of the network structure were used so that users could interact mainly through a network overview rather than only inside the individual documents [13]. In other systems like gIBIS and Aquanet the mapping showed typed links and typed nodes and the types were visually distinguished inside the map view [13]. In these systems, meaning was generally encoded explicitly through predefined link types and structured relationships [13].

Even though link-centered hypertext became the dominant approach, historical reviews show that many users often did not use explicit linking much, even when the tools were available [17]. For example, in retrospective reports about NoteCards, typed links were often underused, and sometimes link-typing was even abandoned during real work [17]. Instead of creating many explicit links, users preferred lighter ways of organizing information, like simple grouping or informal spatial arrangement.

In Anderson's historical analysis he explains that hypertext research in the 1980s and 1990s mainly focused on links, partly since link traversal was technically easier to implement and also matched formal approaches like logic-based structures or argument mapping [17]. However, forcing users to define explicit relationship types too early can create problems. When users are still on exploration process and their understanding is not yet stable, asking them to choose precise link types can slow them down and interrupt their thinking [17]. This is often described as premature formalization, where structure is demanded before the user actually understands the domain well enough [17].

From these observations, spatial hypertext emerged as an alternative to strict link-centered models. Shipman and Marshall describe a move from document-centered hypertext toward map-based interaction, where relationships can be shown not only with explicit links but also through spatial arrangement and visual properties [13]. When information items are placed on a shared workspace or map, users can express meaning through how they position and group items, without immediately creating formal links.

In spatial hypertext, relationships can be expressed implicitly using cues like proximity, alignment, color, shape, or other visual signals, instead of predefined semantic link types [13]. Anderson summarizes this by emphasizing that in spatial hypertext, the relative position and appearance of nodes can already give out meaning, so the explicit links are not always needed for the structure to make sense [17]. At the same time, spatial hypertext does not fully remove links; it simply reduces the dominance of explicit link visualization and allows structure to emerge through layout and interaction [17].

This shift toward spatial hypertext is not only a change in how information is displayed. Shipman and Marshall report that when map-based workspaces were introduced, users naturally started showing relationships through spatial positioning, and in many cases they preferred this implicit method rather than explicit linking [13]. Similar behavior was also seen on how people organize physical documents like paper notes or index cards, which suggests that spatial grouping is a natural strategy humans use when working with information [13].

Because of these findings, researchers argued that hypertext systems should support implicit and transient relationships that appear during active knowledge work, instead of forcing everything into fixed link structures [13]. This idea also matches critiques of link-centered systems that are sometimes described as the "tyranny of the link," meaning links push users to express clear and fixed relationships even when their understanding is still developing [17].

Overall, spatial hypertext can be seen as a conceptual alternative to traditional hypertext because it lets users express meaning through spatial organization before defining formal structures, providing a basis for understanding how spatial cues carry implicit relationships, which is discussed in the next section.

3.1.2 Spatial Cues as Implicit Relationships

In traditional navigational hypertext relationships between information objects/nodes are indicated with typed links that connect nodes in a network [13]. In this structure, meaning is clearly defined within formal connections. A link shows that a relationship exists and also defines its type [13]. This allows precise navigation but it requires users to define relationships clearly. Spatial hypertext changes this approach. Instead of relying mainly on visible linked representations, it allows meaning to emerge from spatial and visual arrangements [13]. Nodes are placed on a shared workspace and their relative position becomes the main way structure is expressed. Shipman and Marshall shows this as supporting implied structure, where relationships based on layout rather than predefined semantic links are understood [13]. In this model, structure is not formally specified in advance. It is derived from the arrangement of objects. This reduces the need to define fixed link types in the early stages of work while still allowing structural interpretation [13]. One of the main spatial cues in spatial hypertext is proximity [12], [13]. Nodes that are placed closer to each other are usually interpreted as related. Nodes that are placed further apart are understood as less related or not connected, depending on user perception [13]. This interpretation does not rely on typed links. Instead, it depends on how users visually interpret spatial distance. In an early system called "VIKI", relationships could be expressed through spatial juxtaposition and placement without directly defining formal links [14]. Users were able to keep relationships implicit in the early stages of work and later formalize them if necessary [14]. This method supports exploratory work. During early sensemaking, users often do not have stable categories or clearly defined conceptual structures [14], [15]. Requiring direct link, typing at this stage may interrupt the thinking process. Spatial placement allows users to express temporary associations without committing to strict strong definitions. Objects can be moved, regrouped, or separated as understanding develops. This flexibility supports gradual clarification of relationships. Over time, some spatial groupings may become more stable and can later be translated into more explicit structures if needed [14]. Because relationships are expressed through spatial configuration rather than predefined link types, interpretation becomes important. Users infer meaning from layout, grouping, and distance. Structure is therefore not imposed by the system but emerges through interaction. This characteristic leads directly to the concept of informal representation, where structure can remain partial, provisional, and open to reinterpretation. The following section examines this aspect in more detail and explains how interpretation by the reader plays a central role in spatial hypertext systems.

3.1.3 Informal Representation and Interpretation by the Reader

Another important aspect of spatial hypertext is that it supports informal representations. User can organize information in meaningful ways without defining explicit link types or fixed metadata in the beginning [13]. Instead of forcing early decision making, spatial hypertext lets the structure stay incomplete, temporary and open to change. This is not a limitation. It reflects how people naturally work while they are still developing their understanding of a topic [13][14]. In link-based hypertext clarity usually comes from making the relationships explicit and often typed/encoded, which motivates for early precision [13]. Spatial hypertext, on the other hand, allows the opposite with underspesification. Items placed near each other, aligned or visaully grouped/clustered can suggest a connection, but the exact meaning of that connection can be left as perspective [13]. VIKI observations show that users often arranged notes to reflect emerging interpretations instead of finalized categories, and these arrangements were repeatedly adjusted as new insights appeared [14]. This makes spatial hypertext particularly suitable for early-stage sensemaking, where relationships are tentative and categories are still evolving [14][15]. Because spatial cues are not fully formal, meaning is not "stored" only in the system; it is also constructed by the reader. Users interpret proximity, clustering, and visual differentiation using contextual knowledge and perceptual inference [13]. This places the user in an active interpretive role: spatial hypertext is readable, but it is not self-explaining in the same way as typed semantic links. Anderson emphasizes that this interpretive quality is central to why spatial hypertext remains valuable—especially when users are exploring, not merely navigating [17]. At the same time, spatial hypertext is not limited to purely human interpretation. A recurring theme in later spatial hypertext work is that informal spatial organization can sometimes be analyzed computationally, for example using parsers that detect clusters and infer relationships from layout [17]. Importantly, Anderson notes that while some communities expect spatial hypertext systems to include spatial parsing, this is not a strict requirement of the paradigm itself [17]. Instead, parsing represents an additional step: translating informal, human-readable structure into machine-usable structure. This point becomes important for the next section. If spatial hypertext supports informal, evolving structure, then an important question is how structure stabilizes over time and how informal organization can transition toward more explicit structure when needed. This transition often described through emergent structure and incremental formalization is the focus of Section 3.2.

3.2 Emergent Structure and Incremental Formalization in Spatial Hypertext

Spatial hypertext does not require users to start with a fixed schema. Instead, it supports a process where organization develops through interaction and gradually becomes clearer. This section explains how structure turns up through spatial work, how it can become more direct over time, and how some systems extend this by interpreting spatial layout computationally.

3.2.1 Emergence Through Interaction and Spatial Arrangement

A foundational principle of spatial hypertext is that structure can emerge through interaction, rather than being fully specified in advance [14]. Users can begin by placing items loosely on a workspace and then refine the arrangement through movement, grouping, separation, and visual differentiation. Over the time repeated interaction creates patterns that become meaningful, clusters stabilize, regions acquire consistent meaning, and some spatial conventions become "rules" through practice rather than system enforcement [14]. This differs from navigational or semantic hypertext, where structure is commonly treated as something that must be explicitly defined to support navigation or reasoning [13]. Spatial hypertext allows users to postpone these decisions. This reduces premature formalization and supports exploration under uncertainty [13][17]. Anderson argues that this flexibility is not just a usability preference it matches the reality of many knowledge tasks, where early structure is unstable and meaning is constructed gradually [17]. This emergent behavior is also consistent with information triage, where users sort and regroup materials before they can name stable categories or relationships [15]. Spatial hypertext supports these early decisions by enabling users to externalize relevance and similarity judgments directly through layout [15]. In this sense, emergence is not randomness; it is the visible trace of evolving interpretation.

3.2.2 From Informal Grouping to More Explicit Structure

Although spatial hypertext starts informally, it does not imply that work must stay informal forever. A key idea in the literature is that spatial organization can support incremental formalization, where users gradually move from loose arrangements toward more explicit structures once meaning stabilizes [13][14]. In practice, this can happen when clusters become compatable enough to be treated as conceptual units: a group might receive a label, a region might become a category-like area, or a set of items might be converted into a collection or hierarchical structure [14]. The important point is that explicit structure becomes a result of interaction rather than a precondition for interaction. The Mother system expresses this idea at an architectural level by describing hypertext as a set of coexisting domains with different degrees of formalization [18]. Mother integrates spatial hypertext with other paradigms (such as node-link and hierarchical forms), and conceptually distinguishes layers such as Midgard (interfaces), Asgard (structure-aware/intelligent services), and Hel (knowledge-related services) [18]. This layered view highlights an important thesis-relevant point: informal spatial workspaces can act as an entry layer for exploration, while more explicit services and representations can be added when required by scale, collaboration, or system-level operations [18]. This layered perspective also clarifies a common tension noted in reflections on spatial hypertext: as workspaces grow, purely informal layouts can become hard to manage, which creates pressure toward additional structure and support mechanisms [17]. Incremental formalization is therefore not only a cognitive process, but also a practical response to growth and complexity [17][18].

3.2.3 Interpreting Spatial Layout Computationally: Parsers and Cluster Detection

A further development, especially important for intelligent and recommender-supported systems, is the idea that spatial organization can be translated into machine-usable structure. While spatial cues are primarily designed for human interpretation [13], systems can also analyze layout to detect clusters and infer relationships computationally. Mother explicitly places such analysis inside its structure-aware services layer, describing specialized parsers (including spatial parsers) that can interpret spatial structures in ways that resemble human recognition of grouping and arrangement [18]. This positions spatial parsing as a bridge between informal human organization and system-level services such as querying, recommendation, and intelligent support. SPORE demonstrates this bridge in a concrete way. In SPORE, users arrange narrative concepts on a spatial canvas, and the system uses a spatial parser to translate the visual layout into a weighted graph representation [19]. By applying thresholds to edge weights, the parser identifies subgraphs that correspond to visual clusters, which then become meaningful units for downstream processing [19]. This is crucial because it shows how spatial hypertext can support computational operations without forcing users to formalize relationships manually. This mechanism directly supports recommendation behavior in SPORE. The system's knowledge base is stored as a weighted graph (managed in Neo4j in the implementation), and recommendations adapt as the user adds or repositions concepts [19]. In other words: the user's spatial actions are not only expressive—they become signals that the system can interpret and respond to. This matches the broader direction highlighted in recent reflections: spatial hypertext remains relevant partly because it can serve as an interface between human sensemaking and intelligent services, including recommender systems and AI-assisted workflows [17].

3.3 Spatial Hypertext as Cognitive Infrastructure for Knowledge Work

While Section 3.2 examined how structure emerges and stabilizes over time, this section focuses on the cognitive role of spatial hypertext. Rather than asking how structure forms, the emphasis here is on how spatial organization supports human reasoning during complex and uncertain tasks. Spatial hypertext does not only represent knowledge but also participates in thinking.

3.3.1 Information Triage: Managing Uncertainty Through Spatial Organization

The concept of information triage describes how users manage large volumes of heterogeneous information under conditions of uncertainty and cognitive overload [15]. Similar to medical triage, before deeper analysis can begin users must make early decisions about what to prioritize, defer, cluster, or discard [15]. Importantly, triage occurs before stable categories exist. Users often do not yet know how information should be structured. Spatial hypertext supports this early stage by allowing provisional grouping through layout rather than requiring formal classification [15]. Research shows that users rely heavily on spatial cues during triage. Items placed close together are interpreted as related; separation indicates distinction; peripheral placement may signal lower priority or unresolved status [15]. These judgments are externalized spatially instead of being stored only in working memory. Triage is iterative. As understanding evolves, users repeatedly move and regroup items [15]. Clusters expand, split, merge, or dissolve. Unlike systems that enforce predefined schemas, spatial hypertext allows this restructuring without breaking formal constraints [17]. This behavior connects directly to the idea of emergent structure described earlier. However, the key contribution here is cognitive: spatial hypertext reduces the burden of early formal decision-making by supporting rapid, revisable grouping under uncertainty [15].

3.3.2 Externalizing Reasoning: Workspace as Persistent Cognitive Support

Apart from grouping and filtering, spatial hypertext can also work as an external memory system. Studies on information triage show that users often move part of their reasoning into spatial arrangement [15]. Instead of keeping tentative relationships in their minds, they express them through placement, clustering, and visual differences. In this way, the workspace becomes part of the thinking process.

One important property of spatial workspaces is persistence. Items remain visible and keep their positions over time [15]. Clusters, alignments, and spatial regions preserve traces of earlier reasoning. Users can return to these traces, interpret them again, and improve them without rebuilding everything from memory.

This persistence makes spatial hypertext different from purely conversational systems. In chat-based interaction, earlier reasoning can become hidden in long message histories. Spatial hypertext, on the other hand, keeps structure visible at the same time rather than burying it in sequence.

The Mother architecture gives a broader view of how spatial workspaces connect to larger knowledge systems [18]. In Mother, informal spatial organization is one domain among others, placed next to more structured hypertext representations [18]. This layered design shows that spatial workspaces can act as a front-end cognitive space, where ideas develop before moving into more formal structures.

The main implication is structural. Spatial hypertext does not only store information. It supports step-by-step reasoning and gradual stabilization. It keeps intermediate stages of understanding visible while still allowing continuous changes.

3.3.3 Benefits and Limits of Informal Spatial Organization

The strengths of spatial hypertext come from its flexibility, interpretability, and its support for exploratory knowledge work. Because relationships are implicit and not formally encoded, users can test different groupings and arrangements without committing to rigid ontologies too early [13]. This helps users tolerate ambiguity and refine their ideas step by step over time [15]. In this way, the workspace can work as a cognitive scaffold, where some reasoning is shared between the user and the spatial environment [15].

At the same time, this informality also brings limitations. First, meaning depends on the reader. Spatial proximity does not have fixed semantics, so different users may interpret the same layout in different ways [13][17]. In collaborative work, this can be difficult, because users may need to discuss and agree on what spatial conventions mean [17]. Second, scalability can become a problem. When a workspace becomes large, informal organization can lead to visual clutter and structural ambiguity [17]. Without extra support, spatial grouping alone may not be enough to manage large collections. Third, spatial hypertext does not automatically provide machine-readable semantics. Spatial layouts help humans understand structure, but automated reasoning usually needs more explicit structure [17]. For this reason, systems such as Mother and SPORE show that extra layers, such as spatial parsers and structured services, may be needed when computational processing becomes important [18][19].

These limitations do not reduce the value of spatial hypertext. Instead, they show where it is most useful. Spatial hypertext is strong for early-stage exploration and for organizing knowledge that is still changing. When tasks need formal inference, large-scale querying, or machine-driven processing, additional mechanisms can be introduced. This leads to the next question. If spatial hypertext provides flexible, persistent, and human-centered structure, then how can it work together with generative AI systems that are probabilistic and mostly use linear chat interfaces? Section 3.4 examines this by analyzing systems that combine spatial organization with AI-supported sensemaking.

3.4 Spatial Workspaces and AI-Supported Sensemaking

Sections 3.1–3.3 established that spatial hypertext supports emergent structure, interpretability, and externalized reasoning. The next step is to examine how these properties interact with large language models (LLMs) and AI-supported knowledge systems. Generative AI systems are commonly accessed through linear chat interfaces. However, sensemaking tasks are rarely linear. They require comparison, restructuring, abstraction, and stabilization across multiple fragments of information. This creates a structural mismatch between chat-based interaction and the spatial organization principles described earlier. This section analyzes how recent systems address this mismatch by embedding AI interaction within spatial or visual workspaces.

3.4.1 Why Linear Chat Interfaces Are Structurally Insufficient for Complex Sensemaking

Large Language Models are typically accessed through sequential chat interfaces, where interaction develops as a growing conversation history [21]. Each new message is added after the previous one, forming a linear timeline. This format works well for simple question–answer exchanges, but complex knowledge work often requires comparing multiple ideas at the same time, revisiting earlier reasoning, grouping related elements, and refining only certain parts without restarting the whole interaction. A linear chat stream makes these activities difficult [21]. As conversations become longer, earlier content can disappear into the history, forcing users to scroll, search, or mentally rebuild structural relationships. This increases cognitive load and reduces structural transparency.

The limitation is not only about navigation but also about structure. Chat-based systems do not naturally preserve spatial grouping, parallel organization, or layered abstraction. Interaction remains mostly sequential even when the thinking process is not. Research on steering LLM summarization shows a similar issue [22]. When users refine summaries through repeated prompts, every step stays inside the same linear stream, and the system does not keep a persistent visual overview of how the structure evolves. Because of this, refinement often depends more on prompt wording than on stable structural context. This structural limitation reflects earlier concerns discussed in Chapter 2 about prompt sensitivity and variability, where small wording changes can lead to large differences in output [9]. Spatial workspaces introduce an alternative structural layer that can address these issues.

3.4.2 Visual Workspaces as Structural Mediation Layers

Recent systems demonstrate that embedding LLM interaction within a visual workspace can address the structural limitations of chat-based interaction.

Sensecape provides a clear example [21]. Instead of presenting generated content only in a chat thread, the system maps concepts into a spatial canvas combined with a hierarchy view. Users can:

- Create nodes
- Group related concepts
- Collapse and expand abstraction levels
- Navigate between overview and detail

This multilevel structure supports simultaneous access to multiple conceptual regions [21]. High-level summaries coexist with mid-level clusters and detailed fragments. Users can move between abstraction levels without losing structural context.

This approach directly reflects earlier spatial hypertext principles. Just as VIKI allowed clusters to stabilize through rearrangement [14], Sensecape allows users to reorganize AI-generated content spatially. Structure is not determined solely by the model. It emerges through user interaction.

Importantly, the workspace is persistent. Unlike chat history, which scrolls out of view, spatial clusters remain visible and manipulable [21]. This persistence reduces the need to mentally reconstruct relationships and supports iterative refinement.

The study reports that participants explored more concepts and revisited prior information more frequently in the multilevel interface than in baseline chat-plus-canvas systems [21]. This suggests that structural externalization improves both breadth and depth of exploration.

From a thesis perspective, the key insight is that the workspace becomes a structural mediator. It does not replace the model. It provides a stable representational environment in which generative output can be reorganized and interpreted.

3.4.3 Spatial Configuration as a Steering Signal for Generative Models

Beyond structural persistence, visual workspaces can also influence generative behavior more directly. Research on steering LLM summarization shows that spatial configuration can function as an input signal for generative systems [22]. In this study, users build a workspace that includes filtered documents, clusters, highlights, and annotations, and this spatial setup is translated into structured prompt content before being sent to the LLM [22]. Within this process, clusters signal conceptual relatedness, selection defines the working scope, filtering reduces noise, and highlighting indicates importance. The workspace therefore acts as an intermediate representational layer between human reasoning and generative processing.

Empirical findings indicate that combining filtering and clustering improves summarization correctness compared to baseline approaches [22], providing quantitative evidence that structural guidance can improve generative alignment. Conceptually, this mechanism is similar to spatial parsing in systems such as SPORE [19], where spatial arrangement is converted into weighted graph structures that guide recommendation and generation. In [22], spatial configuration becomes structured prompt input, while in SPORE it becomes graph-based signals; in both cases, spatial organization gains computational relevance. The distinction is important. In classical spatial hypertext, interpretation is mainly human-driven [13], whereas in AI-supported environments spatial configuration can also guide machine processing. Rather than replacing human interpretation, this creates a feedback loop in which the human organizes spatially and the system responds within the structure that emerges.

3.4.4 Localized Refinement and Iterative Co-Structuring

A further contribution of spatially integrated systems is localized interaction.

In chat-based systems, refining a subset of ideas often requires rewriting the entire prompt. In spatial workspaces, users can select specific clusters and request regeneration or explanation only within that region [21].

This supports localized refinement rather than global restart.

The interaction pattern shifts from:

Prompt → Response → Re-prompt → Response

to:

Generate → Spatially reorganize → Regenerate within selected cluster → Refine

This iterative pattern closely resembles the emergent structure process described in Section 3.2. Structure stabilizes through repeated interaction rather than through single-shot decisions.

This model also aligns with hybrid intelligence theory [8]. The AI contributes generative breadth. The human contributes interpretation and organization. The workspace mediates between them.

Importantly, the workspace preserves conceptual anchors even when generated text changes. Clusters remain visible across iterations. This reduces instability caused by probabilistic output variation [9].

Thus, spatial workspaces do not eliminate uncertainty. They stabilize it.

3.4.5 Structural Implications for Hybrid Intelligence

Taken together, these systems point to three main structural implications. Linear chat interfaces are often not suitable for complex sensemaking because they do not keep structure visible or parallel over time [21]. Visual workspaces, in contrast, can function as mediating layers that preserve structure, support multilevel abstraction, and guide generative output [21][22]. In addition, spatial organization can be translated into computational signals, such as structured prompts or spatial parsing, which allows machine processing without forcing early formalization [18][19][22].

These observations reposition spatial hypertext in relation to generative AI. Rather than being replaced by LLMs, spatial hypertext offers the structural layer that generative models often lack. Generative systems are effective at producing fluent content, but they represent structure in implicit and less transparent ways [3][7], whereas spatial workspaces keep structure visible and persistent. This comparison suggests a clear division of roles: the AI focuses on generating and synthesizing content, the human organizes and interprets it, and the workspace stabilizes and mediates the interaction. This perspective leads directly into Chapter 4, where the structural complementarity between generative AI and spatial hypertext is formalized within a hybrid intelligence framework.

3.5 Synthesis: Structural Contributions of Spatial Hypertext for Hybrid Intelligence

Sections 3.1–3.4 examined spatial hypertext from foundational principles to AI-supported applications. This section consolidates these findings and clarifies their structural implications for knowledge-based systems. Rather than repeating earlier arguments, the focus here is on identifying what spatial hypertext uniquely contributes — and why this contribution becomes critical when combined with generative AI.

3.5.1 Spatial Organization as Interpretable Structure

A central difference between spatial hypertext and link-centered systems is how structure is represented. In traditional navigational or semantic hypertext, structure is defined explicitly through typed links and predefined schemas [13]. Meaning depends on declared relationships. Spatial hypertext, in contrast, represents structure through spatial configuration such as proximity, grouping, alignment, and visual differentiation [13][14]. As a result, relationships become visible in the layout, users can reorganize them dynamically, arrangements remain persistent across interaction, and interpretation stays under human control.

Unlike generative AI systems, where internal representations remain hidden inside model parameters [7], spatial hypertext externalizes structure in a form that users can directly inspect and modify. This interpretability is not only a design choice but also a structural property. In knowledge work, understanding often depends on how fragments relate to each other, not only on the content itself. Spatial organization makes these relationships visible at the level of interaction even when they are not formally encoded. In hybrid human–AI environments, this visible structure becomes especially relevant. Large Language Models can generate fluent text [3], but they do not maintain persistent and inspectable structural scaffolding. Spatial hypertext therefore provides a complementary layer that supports visible and stable organization.

3.5.2 Emergent Structure and Evolving Knowledge

Another important contribution of spatial hypertext is its support for evolving structure. Studies on VIKI and similar systems show that structure develops through repeated interaction instead of being fixed from the beginning [14]. Informal clusters become more stable over time as users adjust and reorganize elements while their understanding grows. This gradual process reflects how knowledge is often built in real-world settings, where categories are not predefined but shaped through exploration, comparison, and reinterpretation [15][17].

This characteristic becomes especially relevant when working with generative AI. Because LLM outputs are probabilistic and can vary across runs [9], knowledge work with these systems tends to be iterative. Users review drafts, compare alternatives, and refine their interpretations step by step. Spatial hypertext offers an external environment where this stabilization can happen. Instead of forcing early formalization, the workspace allows gradual consolidation, where informal clusters can later evolve into more explicit structures when needed [14][18]. In this way, emergent structure acts as a bridge between uncertainty and formal representation, supporting exploratory reasoning while still allowing knowledge to stabilize over time.

3.5.3 Spatial Workspaces as Guidance Mechanisms for AI-Supported Sensemaking

Recent systems show that spatial workspaces do more than support human organization, they can also guide AI processes. Sensecape demonstrates that multilevel spatial exploration allows users to reorganize and refine AI-generated concepts [21], while steering-based summarization systems show that spatial clustering and selection can directly influence LLM outputs [22]. In both cases, the workspace becomes part of the generative context. This leads to an important structural insight: spatial organization can work as a mediating layer between human interpretation and machine generation. Instead of relying only on prompt engineering, users can define scope through selection, show relatedness through clustering, and signal importance through spatial positioning. These spatial signals provide stabilizing context for generative models [22], reduce dependence on fragile prompt wording, and help preserve conceptual anchors across iterations.

From a hybrid intelligence perspective [8], this creates a complementary division of roles in which the AI contributes generative and synthesizing capabilities, the human contributes interpretation and structural judgment, and the workspace mediates between them by providing visible and persistent structure. This interaction pattern shifts generative AI from being only a conversational interface toward becoming part of a structured sensemaking environment. The literature reviewed in this chapter highlights three main structural contributions of spatial hypertext: it offers interpretable spatial structure that makes relationships visible [13][14], supports emergent and evolving organization that enables gradual stabilization of knowledge [14][17], and enables workspace-mediated interaction that can guide and constrain AI-supported generation [21][22]. Spatial hypertext does not replace generative AI, and generative AI does not make spatial hypertext unnecessary. Instead, they address different structural needs. Generative AI provides generative breadth and fast synthesis but lacks persistent and inspectable structure [7][9], while spatial hypertext offers persistent, human-controlled organization but does not provide automated generation or large-scale synthesis [17]. This structural asymmetry motivates the integration discussed in the next chapter, where Chapter 4 moves from descriptive analysis toward a conceptual formulation that explains how generative AI and spatial hypertext can be combined within a hybrid intelligence framework while preserving human-centered structural control and using computational generativity.

Chapter 4

Toward a Hybrid Intelligence Framework for Knowledge-Based Systems

4.1 Complementary Strengths and Structural Gaps

Chapters 2 and 3 talked about two different ways of supporting knowledge work. Chapter 2 focused on generative AI systems built on Large Language Models and chapter 3 focused on spatial hypertext systems including VIKI, the Mother architecture, and systems such as SPORE that add computational services to spatial workspaces. Even though both paradigms support knowledge-intensive tasks, they follow different structural principles. It is important to understand these differences before introducing a hybrid intelligence framework.

4.1.1 Generative AI: Generative Breadth Without Persistent Structure

As discussed earlier in Chapter 2, Large Language Models generate natural language by modeling probability distributions over token sequences [3]. They support drafting, summarization, explanation, and exploratory reasoning [3]. AI copilots show that such models can be embedded into writing and programming workflows [5]. They can also process and synthesize large volumes of text efficiently [7].

These capabilities make generative AI useful for expanding ideas and producing candidate knowledge artifacts. During early ideation, LLMs can quickly generate alternatives, summaries, and reformulations but their structural properties differ from traditional knowledge systems. First, model outputs are probabilistic. Identical prompts may produce different responses because generation involves sampling [3, 9]. This variability does not prevent usefulness, but it reduces structural stability. Generated text is not anchored to persistent conceptual structures unless external mechanisms are introduced. Second, knowledge inside the model is implicit. It is encoded in distributed parameters rather than explicit entities or relations [7]. The system can produce coherent explanations, but it does not expose inspectable structural representations. The reasoning process remains opaque. Third, interaction is typically sequential. Chat-based systems organize knowledge as dialogue history. Earlier outputs may become difficult to revisit or reorganize structurally. Context window limits constrain how much prior material can influence new generation [9]. When tasks become complex, this can lead to fragmentation. Overall, generative AI provides generative breadth and rapid synthesis. It adds to conceptual space efficiently, but it does not inherently provide persistent, inspectable, or reorganizable structure.

4.1.2 Spatial Hypertext: Persistent Organization Without Generative Expansion

Chapter 3 showed that spatial hypertext organizes knowledge differently. Instead of typed links, it uses spatial and visual cues to represent relationships [13]. Proximity, grouping, and alignment allow users to express how knowledge pieces are in an implicit way.

In systems such as VIKI, users arrange notes freely in a two-dimensional workspace [14]. Through repeated movement and regrouping, clusters emerge. Structure stabilizes gradually through interaction rather than being predefined [14, 15]. This supports incremental formalization and preserves interpretability.

The Mother system extended this idea by introducing a layered architecture [18]. Informal spatial organization could coexist with more structured hypertext services. This separation allowed flexible grouping at the workspace level while enabling structured processing underneath.

SPORE further demonstrated how spatial arrangements can be translated into computational representations [19]. Its spatial parser converts node configurations into weighted graph structures. In this way, implicit spatial groupings become machine-usable signals without removing user control over layout.

These systems show that spatial hypertext is not only a visual interface.

It can function as a structural mediation layer between informal human organization and computational services.

However, spatial hypertext also has limits. It does not generate new content automatically. Users must create or import nodes manually. Without integration with generative systems, expansion remains limited by human effort. In addition, large workspaces can become difficult to manage without additional indexing or parsing mechanisms [17].

In summary, spatial hypertext provides persistent, visible, and user-controlled structure. It stabilizes evolving knowledge. However, it does not provide generative breadth.

4.1.3 Structural Asymmetry and the Need for Integration

When generative AI and spatial hypertext are examined together, a structural asymmetry becomes visible.

Generative AI compresses knowledge into internal parameter space and produces transient text sequences. Structure is implicit and model-internal.

Spatial hypertext externalizes knowledge into visible spatial configurations. Structure is explicit, user-interpretable, and persistent.

This difference creates a level mismatch:

- Generative AI operates at the level of token probabilities.
- Spatial hypertext operates at the level of conceptual groupings.

Generative systems optimize for linguistic coherence. Spatial systems optimize for conceptual organization.

This asymmetry explains why neither paradigm alone fully supports complex and evolving knowledge work.

Pure generative interaction risks drift and loss of structural overview because interaction is sequential and probabilistic [9]. Pure spatial interaction requires manual effort for content expansion and lacks automated synthesis [17].

Hybrid intelligence research argues that effective systems distribute cognitive labor between human and machine according to complementary strengths [8]. In this context:

- Generative AI expands and synthesizes.
- The human interprets and evaluates.

- The spatial workspace stabilizes and externalizes structure.

Existing systems illustrate partial realizations of this pattern. Sensecape maps LLM-generated content into multilevel spatial canvases that support exploration across abstraction levels [21]. SPORE translates spatial groupings into structured queries through parsing mechanisms [19]. Space-steered summarization shows that visual grouping can guide generative output [22].

These examples demonstrate that integration is structurally motivated. Generative AI requires external stabilization mechanisms. Spatial hypertext benefits from generative expansion and computational assistance.

To summarize this asymmetry, Table 4.1 compares the two paradigms across core structural dimensions.

Table 4.1: Structural asymmetry between generative AI and spatial hypertext in knowledge work.

Dimension	**Generative AI (LLMs)**	**Spatial Hypertext**
Representation of structure	Implicit, encoded in model parameters [7].	Explicit, expressed through spatial arrangement [13].
Stability across iterations	Probabilistic and variable [9].	Persistent and visible in the workspace [14].
Interaction mode	Sequential dialogue history.	Spatial and reorganizable layout.
Primary strength	Rapid synthesis and expansion [3].	Human-centered stabilization and interpretation [15].
Primary limitation	Lack of inspectable and persistent structure.	Lack of automated content generation.

The structural gap identified in this section defines the problem space for the remainder of the chapter. The next section examines how existing systems attempt to bridge this asymmetry through interaction mechanisms between human, AI, and workspace.

4.2 Interaction Patterns Between Human, AI, and Workspace

Section 4.1 identified a structural asymmetry between generative AI and spatial hypertext. Generative systems provide probabilistic expansion without persistent structure. Spatial systems provide persistent structure without automated expansion.

This section examines how existing systems attempt to bridge this asymmetry through interaction patterns. The focus is not on full architectural design, but on recurring mechanisms observed in AI-supported spatial workspaces.

4.2.1 The Human–AI–Workspace Interaction Loop

Traditional interaction with generative AI is linear. A user submits a prompt, and the model produces a response. The process unfolds as sequential dialogue. Earlier outputs move upward in the conversation history and may fall outside the active context window [9]. Structural overview is limited.

When spatial hypertext is introduced, interaction changes. The system becomes triadic:

- The human agent
- The generative AI model
- The spatial workspace

The workspace is not only a display surface. It becomes a structural medium that stores and organizes intermediate results.

In Sensecape, LLM-generated content appears as nodes on an infinite canvas [21]. Users group nodes, create sub-canvases, and navigate between abstraction levels. Interaction no longer depends only on prompt history. The canvas preserves conceptual relationships visually.

In SPORE, narrative elements are arranged spatially by the author [19]. The generative component produces prose based on spatially grouped elements. The workspace mediates between generation and interpretation.

Across these systems, a recurring interaction loop can be observed:

1. The model generates candidate artifacts.
2. The user reorganizes them spatially.
3. The spatial configuration defines a new scope of attention.
4. The model generates or refines content within that scope.

This loop transforms linear prompt–response interaction into iterative co-structuring. The workspace preserves structure across iterations, reducing drift caused by probabilistic variability [9]. Even when text changes, spatial anchors remain visible.

In this process, structural authority remains with the human user. The model proposes. The user organizes. The workspace stabilizes.

4.2.2 Structural Mediation: Translating Space into Computation

Spatial hypertext expresses relationships implicitly through layout [13], while generative AI systems operate on computational representations and therefore require structured context [3]. Connecting these two logics requires a mediation layer. Existing systems implement this mediation in different ways, but they follow similar ideas. In SPORE, a spatial parser analyzes node positions and builds weighted graph structures [19]. Clusters are detected based on proximity and edge weights, and these clusters can then be translated into structured queries, which makes implicit spatial grouping usable for machines without removing user control. In space-steered summarization, the workspace is converted into structured prompt components [22]. Selected clusters, highlights, and filters define the scope of summarization, meaning that the spatial configuration becomes part of the input context. Sensecape follows a related approach, where users select specific nodes or sub-canvases to trigger localized generation [21]. Even though this mechanism is not always described as graph parsing, it follows the same principle because spatial selection defines the computational scope.

Across these systems, mediation usually follows a shared process: the spatial configuration is first interpreted, then selected structures are translated into structured input, and finally generation is constrained by that structure. Importantly, this mediation does not require full ontology encoding. Clusters can remain informal and provisional, which aligns with the idea of incremental formalization discussed in Chapter 3 [14, 18]. In this way, structure emerges gradually through interaction before becoming fully formal. Table 4.2 summarizes typical workspace-derived signals that can guide AI operations.

These signals show how spatial organization can function as contextual input for generative systems without enforcing rigid semantic models.

4.2.3 Limits of Current Interaction Mechanisms

Although existing systems show promising integration patterns, several limitations still remain. One issue is that spatial parsing mechanisms are often

Table 4.2: Workspace-derived signals that can guide generative AI operations.

Workspace signal	Meaning in interaction	Evidence
Selection / focus region	Defines the scope of generation or summarization.	[22]
Clustering / grouping	Indicates conceptual relatedness and supports structured output.	[22]
Cluster labels	Adds user-defined semantics that frame generation.	[22]
Hierarchical levels	Supports multilevel exploration and abstraction control.	[21]
Spatial proximity	Suggests conceptual similarity without explicit links.	[13]

domain-specific. For example, SPORE's weighted graph extraction is mainly designed for narrative domains [19], and more general spatial interpretation for open-ended knowledge work is still limited. Another limitation is that many systems support localized generation but do not maintain a persistent computational model of the whole workspace structure. In most cases, the AI reacts only to selected regions instead of reasoning over the complete structural configuration [21]. In addition, translation between the workspace and the AI is often one-directional. The spatial workspace guides the model, but the model itself does not directly reorganize spatial structure, which means that human intervention continues to play a central role in shaping the workspace.

These limitations do not weaken the hybrid approach. Instead, they point to important architectural challenges that need to be addressed by a more generalized hybrid intelligence framework. The next section brings these observations together and identifies the structural conditions that future hybrid systems should satisfy.

4.3 Structural Conditions for Hybrid Intelligence

Sections 4.1 and 4.2 examined the structural asymmetry between generative AI and spatial hypertext and analyzed how existing systems try to bridge this gap through interaction mechanisms. This section brings these observations together. The aim is not to introduce a concrete architecture yet, but to clarify the structural conditions that any hybrid intelligence framework needs in order to support knowledge-based systems effectively.

4.3.1 Condition 1: Persistent External Structure

Chapter 2 showed that generative AI produces probabilistic and sequential outputs [9]. Interaction usually follows a linear dialogue format, where earlier reasoning can become harder to retrieve or reorganize as conversations grow longer, which makes structural continuity fragile. In contrast, Chapter 3 demonstrated that spatial hypertext provides persistent external organization [13, 14]. Clusters remain visible and can be modified without regenerating content, and emergent structure stabilizes gradually through ongoing interaction [14, 15]. The interaction patterns talked about in Section 4.2 supports this difference. In Sensecape, canvases preserve exploration paths [21]; in SPORE, spatial configuration influences later prose generation [19]; and in space-steered summarization, clustering improves alignment between user intent and model output [22]. Taken together, these observations suggest that hybrid systems need a persistent external structure that can store intermediate states, keep conceptual groupings across iterations, and allow reorganization without depending on generative variability. Without this type of persistence, probabilistic generation can easily lead to fragmentation and loss of structural coherence.

4.3.2 Condition 2: Human-Centered Structural Authority

Generative AI provides expansion and synthesis [3, 7], but it does not expose inspectable structural reasoning [7]. Outputs remain provisional and may vary across runs [9].

Spatial hypertext research emphasizes that structure emerges through user interaction [13, 14]. Even in systems with computational assistance, such as SPORE, users decide how nodes are arranged and which suggestions to accept [19].

The interaction loop described in Section 4.2 reinforces this principle. The model generates candidate artifacts. The user reorganizes them spatially. The workspace stabilizes interpretation. Structural authority remains with the human agent.

This implies that hybrid systems must preserve human-centered structural control. Generative processes should operate within boundaries defined by user interpretation and workspace organization. The model expands possibilities, but it does not autonomously determine conceptual structure.

4.3.3 Condition 3: Mediation Between Informal Space and Computational Context

Spatial hypertext expresses relationships implicitly through layout [13]. Generative AI systems require structured input to guide output [3]. Bridging these logics requires mediation.

Section 4.2 showed that mediation can occur through spatial parsers [19], structured prompt translation [22], or localized node selection [21]. These mechanisms convert spatial configuration into computationally usable signals without enforcing rigid ontology encoding.

Chapter 3 emphasized incremental formalization [14, 18]. Structure should emerge through interaction before becoming fully formal. Mother demonstrated that informal spatial domains can coexist with more structured processing layers [18].

Therefore, hybrid systems require a mediating layer that:

- interprets spatial configuration,
- translates selected structures into computational scope,
- preserves informality where appropriate,
- avoids premature semantic rigidity.

Mediation must balance flexibility and machine usability. It should extract structural signals while preserving the evolving nature of knowledge organization.

4.3.4 Condition 4: Iterative Co-Structuring

Generative AI interaction is often treated as single prompt–response exchange. However, Chapter 2 highlighted the instability of relying on isolated outputs [9].

Spatial hypertext research showed that knowledge stabilizes through repeated rearrangement and refinement [14, 15]. Structure evolves gradually rather than being fixed at creation.

The systems analyzed in Section 4.2 follow a recurring pattern:

1. Generation of candidate content.

2. Spatial reorganization by the user.

3. Localized regeneration or refinement.

This iterative co-structuring reduces drift and supports gradual stabilization. It transforms knowledge work from linear response production into structural evolution.

Hybrid systems must therefore support interaction cycles rather than isolated generation events. Generation should enter a spatial environment where interpretation, reorganization, and refinement can occur repeatedly.

4.3.5 Synthesis

The analysis across Chapters 2, 3, and 4 reveals a clear structural landscape.

Generative AI contributes probabilistic expansion and synthesis but lacks persistent and inspectable organization. Spatial hypertext contributes emergent and user-controlled structure but lacks generative breadth and automated synthesis.

Existing systems demonstrate partial integrations through spatial parsing, multilevel canvases, and workspace-guided summarization. However, no unified framework systematically integrates these mechanisms into a general hybrid intelligence model.

The four structural conditions identified in this section define the conceptual foundation for such a model:

- Persistent external structure,

- Human-centered structural authority,

- Mediation between informal space and computational context,

- Iterative co-structuring.

These conditions do not yet describe a specific architecture. Instead, they define the problem constraints that a hybrid intelligence framework must satisfy.

The next chapter builds on these conditions and formalizes them into explicit principles and a conceptual architecture for integrating generative AI and spatial hypertext in knowledge-based systems.

Chapter 5

A Hybrid Intelligent Framework

5.1 Principles of Hybrid Intelligence

Chapter 4 identified four structural conditions for integrating generative AI and spatial hypertext: persistence of structure, human-centered authority, mediation between informal space and computational context, and iterative co-structuring.

This section translates these structural conditions into explicit design principles. While Chapter 4 analyzed what is required for integration, this section defines how a hybrid intelligence system should be organized in order to satisfy those requirements.

5.1.1 Principle 1: Human-Centered Structural Control

The first principle is that structural control should remain with the human user. The Generative AI can generate drafts, summaries, and oprtional formulations, but it should not autonomously decide conceptual organization.

Research on Generative Artificial Experts highlights bounded autonomy, where AI systems support complex tasks while remaining under human supervision [6]. Studies on generative AI in knowledge work similarly stress that liability and explanation cannot be fully delegated to models [7]. Hybrid intelligence is therefore understood as a complementary collaboration rather than a substitution of human reasoning [8].

Spatial hypertext provides a structural model that supports this position. In VIKI, structure emerges through user-driven clustering and rearrangement rather than through predefined schemas [14]. Earlier the critiques of link centered hypertext had caution against premature formalization, where rigid structures are forced before understanding matures [13, 17]. These findings suggest that conceptual organization should remain under user control even when computational assistance is available.

In the proposed framework, the AI module produces candidate knowledge artifacts. The user evaluates these artifacts and determines how they are grouped, prioritized, or refined within the workspace. The spatial workspace becomes the site where structure is stabilized. In this way, generative expansion supports human reasoning without overriding it.

This principle directly refers to the Research Question 1 by defining the division of roles between human and AI within the hybrid systems.

5.1.2 Principle 2: Emergent-to-Formal Transition

The second principle is that the hybrid systems must support a slow transition from informal organization to more explicit structure. Early stage knowledge work often involves unpredictability, vagueness, and incomplete understanding. Forcing firm schemas at this stage can distort evolving interpretation.

Spatial hypertext research always shows that structure emerges through an interaction. In VIKI, users at first place items loosely and later recognize stable clusters as understanding develops [14]. Information triage studies demonstrate that regrouping precede stable categorization and sorting [15]. Structure is therefore a product of interaction, not a predefined constraint.

The Mother architecture further illustrates how informal spatial organization can coexist with more structured domains [18]. Informal groupings may later inform structured representations, but formalization is layered rather than immediate.

Generative AI outputs reinforce the need for this principle. LLM responses are probabilistic and may vary across runs [9]. In addition, knowledge encoded in model parameters is implicit and not directly inspectable [7]. AI-generated content should therefore be treated as provisional material that enters the workspace for comparison, grouping, and revision.

In the framework, AI-generated elements first exist in an informal spatial state. As clusters stabilize through recurrent interaction, optional formalization may occur, such as labeling groups or defining clear relations. Formal structure becomes an outcome of stabilization rather than a prerequisite for exploration.

This principle directly addresses Research Question 2 by clarifying the role of emergent and informal representation in hybrid intelligence.

5.1.3 Principle 3: Workspace-Mediated AI Guidance

The third principle is that the spatial workspace should function as an active guidance layer for generative AI. The Hybrid systems should not depend solely on textual prompts to control model behavior. Instead, spatial configuration must contribute to computational context.

Visual steering research demonstrates that grouping, selection, and arrangement within a workspace can influence LLM summarization output [22]. When users select clusters or define regions of focus, these spatial actions determine what the model emphasizes. Similarly, Sensecape shows that generation can be localized to selected nodes or abstraction levels within a spatial canvas [21].

Classical spatial hypertext theory explains why this is possible. Spatial cues such as proximity and grouping express implicit relationships [13]. When these cues are translated into computational signals—through selection, clustering, or parsing—they can serve as structured input without requiring rigid semantic encoding [17].

In the proposed framework, AI operations are scoped by workspace structure. Clusters, labels, hierarchical layers, and selected regions define what content the model processes and how output is generated. This decreases reliance on a persistent and interpretable control mechanism on prompt wording alone.

This principle directly addresses Research Question 3 by defining how the workspace mediates between human reasoning and generative computation.

5.1.4 Principle 4: Iterative Co-Structuring as the Default Interaction Mode

The fourth principle states that hybrid intelligence systems should work through iterative co-structuring instead of isolated prompt–response exchanges.

Chapter 4 showed that AI-supported spatial systems often follow a repeating interaction pattern: the model generates candidate artifacts, the user reorganizes them spatially, and the updated structure guides the next generation step [21, 22]. This process is consistent with spatial hypertext research, where structure becomes more stable through repeated rearrangement [14, 15].

Because LLM outputs are probabilistic, single-step generation is usually not enough for complex knowledge tasks [9]. Iteration allows users to compare alternatives, refine clusters, and slowly stabilize meaning through persistent spatial anchors.

Within the framework, iteration is treated as the normal way of working.

AI output is considered provisional rather than final. The workspace keeps intermediate structure visible across cycles. Through repeated rounds of generation and reorganization, knowledge develops from exploratory fragments into more stable conceptual structure.

Together, these four principles form the normative foundation of the hybrid intelligence framework.

5.2 Conceptual Architecture of the Hybrid Intelligence Framework

The principles introduced in Section 5.1 explain how hybrid intelligence systems are expected to operate. This section turns those principles into a conceptual architecture. The aim is not to describe a specific software implementation, but to outline the main structural components and the interaction logic needed for integration.

The architecture includes four connected components: a human agent, a spatial hypertext workspace, a generative AI module, and a mediation layer that converts spatial organization into computational context. Together, these components reflect the structural conditions discussed in Chapter 4.

5.2.1 Core Components

Human agent. The human user remains the central interpretive authority in the system. Hybrid intelligence is defined as a collaboration in which human reasoning and machine computation are complementary rather than substitutive [8]. In spatial hypertext, meaning is formed through user interaction with layout and grouping [13, 14]. Within the framework, the human evaluates AI-generated artifacts, organizes them spatially, and decides when structure is sufficiently stable. Structural authority therefore remains human-centered.

Generative AI module. The generative module provides drafting, summarization, synthesis, and alternative formulations [3, 5]. Its outputs are probabilistic and context-dependent [9]. The module does not define permanent structure. Instead, it produces candidate artifacts that enter the workspace for evaluation and reorganization. In this role, the AI functions as a proposal generator rather than as a structural decision-maker.

Spatial hypertext workspace. The workspace provides persistent and visible organization. Relationships are expressed through proximity, visual arrangement and grouping,rather than only fixed typed links [13]. Structure stabilizes through clustering and repeated rearrangement [14]. The workspace also supports cognitive offloading by externalizing evolving interpretation [15]. It functions as shared representational space where both human and AI-generated elements become part of a structured environment.

Mediation layer. To make spatial organization computationally usable, a mediation mechanism is required. Prior systems demonstrate different forms of mediation, such as spatial parsing in SPORE [19] and workspace-to-prompt translation in visual steering systems [22]. In the proposed framework, the mediation layer converts workspace signals—such as selection, clustering, labeling, and hierarchy—into structured context that constrains AI operations. This translation preserves informality while enabling computational guidance.

These four components altogeather implement the principles defined earlier: human-centered control, workspace-mediated guidance, iterative co-structuring and emergent-to-formal transition.

5.2.2 Interaction Logic: The Iterative Co-Structuring Cycle

The main behavior of the architecture is a repetitive interaction cycle. Instead of treating AI output as final, the system allows users to improve results step by step through spatial organization.

The cycle includes five stages:

1. **Generate.** The AI module creates candidate artifacts based on the current user goal and the selected workspace context.

2. **Externalize.** The generated artifacts are placed into the workspace as editable elements, such as nodes or notes.

3. **Organize.** The user groups, moves, labels, or separates elements to show their developing understanding.

4. **Translate.** The mediation layer transforms the chosen spatial configuration into structured computational context.

5. **Refine.** The AI generates new output again, now guided by the updated workspace structure.

This cycle reflects interaction patterns described in AI-supported spatial systems [21, 22]. It also helps deal with the variability of probabilistic generation [9]. Instead of relying on fully predictable model behavior, structure becomes more stable through repeated spatial adjustments.

An important aspect is that the workspace keeps intermediate states across cycles. Even when generated text changes, spatial clusters and arrangements stay visible. This supports continuity and helps users follow the development of ideas in complex knowledge tasks.

5.2.3 Gradual Stabilization and Optional Formalization

The architecture does not assume that knowledge begins in a formal representation. At first in a task, clusters and relationships may remain provisional. This aligns with spatial hypertext research, which emphasizes the value of ambiguity and flexibility during early exploration [17].

As interaction continues, certain clusters may stabilize. Stabilization happens when users repeatedly group the same elements together, maintaining spatial proximity, or introducing consistent labels. At this point, optional formalization may occur. For example, stabilized clusters may be assigned explicit names and transformed into typed relations or exported into structured representations.

The Mother architecture provides a precedent for layered coexistence of informal and structured domains [18]. Informal spatial organization can precede more formal representations without requiring immediate ontology commitment.

In the proposed framework, formalization is therefore a consequence of interaction rather than a starting requirement. This keeps exploratory flexibility mean while allowing consolidation when conceptual clarity increases.

Altogether, the core components and interaction logic define a conceptual but structurally grounded model for integrating generative AI and spatial hypertext in knowledge-based systems.

5.3 Distinguishing Features of the Proposed Framework

Several recent systems show how generative AI and spatial workspaces can be combined, although the integration is still not complete. Sensecape brings LLM-generated content into a multilevel canvas environment [21]. SPORE connects spatial hypertext with recommender and generative components for narrative co-creation [19]. Visual steering approaches also demonstrate that spatial grouping and user selection can influence how LLMs produce summaries [22]. Together, these systems give useful empirical examples showing that spatial organization and generative mechanisms can work together.

However, most of these systems focus on specific application areas or on single interaction techniques. The framework proposed in this thesis takes a different approach by describing integration as a more general structural model. Its main differences can be explained through three dimensions.

5.3.1 Generalized Structural Mediation

Recent systems show partial integration between generative AI and spatial workspaces. Sensecape brings LLM-generated content into a multilevel canvas environment [21]. SPORE combines spatial hypertext with recommender and generative components for narrative co-creation [19]. Visual steering approaches also show that spatial grouping and selection can influence how LLMs generate summaries [22]. Together, these systems provide practical examples that spatial and generative mechanisms can work together.

However, most of these approaches focus on specific domains or on individual interaction techniques. The framework proposed in this thesis takes a different direction by describing integration as a more general structural model. Its main dif

5.3.2 Iterative Co-Structuring as the Core Interaction Logic

Existing systems often support recurrent refinement, but these repetitons are treated as a feature rather than as the defining structural principle. For example, Sensecape enables localized regeneration within selected spatial regions [21], and [22] demonstrates improved summarization after workspace organization [22].

In the proposed framework, the iterative co-structuring is the primary interaction model. The system is designed around repeated cycles of generation, spatial organization, translation, and refinement. This shifts the focus from isolated task assistance to sustained structural evolution in knowledge work.

Iteration is therefore not an optional enhancement but the default mode of operation. The framework assumes that knowledge stabilizes gradually through interaction rather than through single-step output.

5.3.3 Explicit Integration of Informality and Optional Formalization

Spatial hypertext research emphasizes emergent and informal structure [14, 17], and the Mother architecture demonstrates layered coexistence of informal and structured domains [18]. However, generative AI integrations do not always explicitly preserve this transition.

The proposed framework formally incorporates the emergent-to-formal transition as a design principle. Early-stage organization remains informal and flexible. Stabilization occurs through repeated spatial reinforcement. Formalization is optional and follows conceptual maturation rather than preceding it.

This plain protection of informality is particularly important in knowledge-based systems, where premature encoding into rigid schemas can limit exploration.

Collectively, these distinguishing features position the proposed framework as a generalized structural synthesis. It combines Generative AI and spatial hypertext at the level of architectural principles rather than at the level of isolated features or domain-specific implementations.

5.4 Research Contribution and Positioning

The hybrid intelligence framework proposed in this chapter contributes to the literature in three interrelated ways.

5.4.1 Structural Synthesis Across Research Domains

First, the framework brings together two research traditions that have mostly developed separately: spatial hypertext and Generative AI. Research on Gen-AI mainly focuses on probabilistic generation, model capabilities, and prompt-based interaction [3, 7]. Spatial hypertext research, in contrast, focuses on emergent structure, implicit organization, and human-centered explainability [13, 14].

Although some recent systems show partial integration of these ideas [19, 21, 22], a clear theoretical model that explains how both paradigms can be combined in a systematic way has been limited. The proposed framework addresses this gap by identifying complementary structural roles and by describing how these elements can interact.

In this way, the thesis goes beyond simple feature integration and presents a structured architectural model grounded in existing research.

5.4.2 Explicit Design Principles for Hybrid Knowledge Systems

Secondly, the framework formulates the four explicit design principles derived from the structural analysis in Chapters 2–4:

- Human-centered structural control,
- Emergent-to-formal transition,
- Workspace-mediated AI guidance,
- Iterative co-structuring.

These principles implement the structural conditions identified earlier. They transform descriptive observations into normative guidance for system design. Rather than proposing a particular interface or algorithm, the framework predefines constraints and roles that the hybrid systems should satisfy in order to support interpretable and structured knowledge work.

This contribution responds directly to Research Question 1 by clarifying how generative AI and spatial hypertext can be combined in a principled manner.

5.4.3 Conceptual Architecture for Human–AI–Workspace Collaboration

Third, the framework introduces a conceptual architecture that separates human agency, generative computation, spatial organization, and mediation into distinct but interdependent components. This layered model clarifies how:

- human judgment maintains structural authority,
- generative AI provides exploratory expansion,
- spatial workspaces stabilize evolving interpretation,
- mediation mechanisms translate informal structure into computational context.

By making these roles clear, the framework contributes to understanding spatial hypertext as a shared cognitive workspace between humans and Generative AI. This directly addresses Research Question 3.

Importantly, the framework does not claim empirical validation. It is a theory-building contribution grounded in systematic literature analysis. Its purpose is to clarify design logic and provide evaluative criteria for analyzing existing and future systems.

The next chapter applies this framework with an analytical lens. By examining representative systems through the four principles and architectural components defined here, it evaluates how current implementations realize or fail to overlook key aspects of hybrid intelligence.

5.5 Chapter Summary

This chapter also translates the structural analysis from Chapter 4 into a formal hybrid intelligent framework. It defined four core principles: human-centered structural control, workspace-mediated AI guidance, iterative co-structuring and emergent-to-formal transition. These principles articulate how generative AI and spatial hypertext can be combined in a way that preserves interpretability while enabling computational expansion.

Based on these principles, the chapter introduced a conceptual architecture consisting of a human agent, a generative AI module, a spatial hypertext workspace, and a mediation layer that connects spatial organization to computational context. The framework defines clear roles for each component and emphasizes iterative interaction as the central mechanism of knowledge development.

Unlike domain-specific integrations, the proposed framework operates at a structural level. It does not prescribe a particular implementation. Instead, it provides design constraints and analytical criteria for evaluating hybrid knowledge systems.

The next chapter applies this framework as an evaluative lens. By examining existing systems in light of the defined principles and architectural components, it assesses how current implementations realize hybrid intelligence and where important gaps remain.

Chapter 6

Framework Application: Analysis of Existing Hybrid Systems

6.1 Analytical Approach and Evaluation Criteria

Chapter 5 introduced a theorethical hybrid intelligence framework based on four principles including a conceptual architecture that includes a human agent, a generative AI module, a spatial hypertext workspace, and a mediation layer. This chapter uses that framework as an analytical lens to examine selected existing systems. The purpose is to analyze how current implementations realize or fail to realize the structural conditions defined earlier. Each system is therefore examined in relation to both the integration principles and the architectural components.

6.1.1 From Principles to Analytical Indicators

For a structured comparison, the four principles from Chapter 5 are translated into observable indicators within the analysis. The first principle, human-centered structural control (P1), examines whether users keep authority over conceptual grouping and organization or whether structure is imposed automatically by the system. The second principle, emergent-to-formal transition (P2), considers whether systems allow informal organization that can gradually stabilize over time, or whether structure is predefined from the beginning. The third principle, workspace-mediated AI guidance (P3), evaluates whether spatial or structural organization is transformed into computational context that influences AI behavior. The fourth principle, iterative co-structuring (P4), investigates whether systems support repeated cycles of generation and spatial reorganization, or whether interaction remains mostly linear.

In addition to these principles, the architectural model highlights the importance of mediation mechanisms. For this reason, a fifth structural dimension is included in the analysis: the mediation layer. This dimension asks whether an explicit mechanism exists that translates spatial configuration into machine-usable representations, such as spatial parsing, structured prompt encoding, or graph extraction. Together, these dimensions make it possible to compare systems consistently even when they differ in domain, purpose, or technical implementation.

6.1.2 Comparative Evaluation Rubric

Each system is assessed qualitatively along the above dimensions using three levels:

- **Strong** — The principle is explicitly implemented and central to system design.
- **Partial** — The principle is present but limited in scope or depth.
- **Limited** — The principle is weakly realized or largely absent.

Table 6.1 summarizes the comparative assessment that will be justified in the following subsections.

Table 6.1: Comparative assessment of selected systems through the hybrid intelligence framework.

System	P1	P2	P3	P4	Mediation
SPORE [19]	Strong	Partial	Strong	Partial	Strong
Sensecape [21]	Strong	Strong	Partial	Strong	Limited
[22]	Strong	Partial	Strong	Partial	Strong
EDGE [23]	Partial	Limited	Limited	Limited	Strong (schema-based)
AISSH [?]	Strong	Partial	Partial	Partial	Limited

The classifications in Table 6.1 are not numerical evaluations but structured interpretations derived from system descriptions and reported functionalities. The following sections examine each system individually and justify these assessments with reference to their architectural design and interaction mechanisms.

6.1.3 Scope and Interpretation

The selected systems show different ways of combining spatial approaches with generative AI. SPORE combines spatial hypertext with recommendation features and generative text production in narrative domains [19]. Sensecape introduces multilevel spatial exploration to support LLM-based sensemaking [21]. [22] provides empirical results showing that visual workspaces can guide summarization processes, while EDGE [23] connects conversational AI with formally structured knowledge graphs. AISSH represents an exploratory prototype that links a spatial hypertext interface with generative chat functionality [24]. Together, these examples include emergent spatial systems, ontology-driven designs, and early prototype integrations. Looking at them through one shared perspective helps reveal common integration patterns and also highlights existing structural gaps. The next section starts with SPORE, since it offers one of the clearest examples of spatial parsing and workspace-mediated AI guidance.

6.2 Case Analysis

6.2.1 Case Analysis I: SPORE

SPORE [19] represents one of the most explicit integrations of spatial hypertext with structured knowledge services and generative components. Rather than describing the system in isolation, this subsection evaluates it through the hybrid intelligence framework defined in Chapter 5.

Human-Centered Structural Control (P1)

SPORE strongly realizes human-centered structural control. Authors arrange narrative elements—such as characters, themes, and objects—directly within a spatial hypertext canvas. As visible in Figure 6.1, story elements are positioned freely, clustered visually, and reorganized interactively. The system proposes suggestions, but it does not impose structure. Users decide which concepts to include, how to group them, and which suggestions to accept.

This aligns directly with Principle 1. Structural authority remains with the human author. The system augments ideation but does not override conceptual grouping.

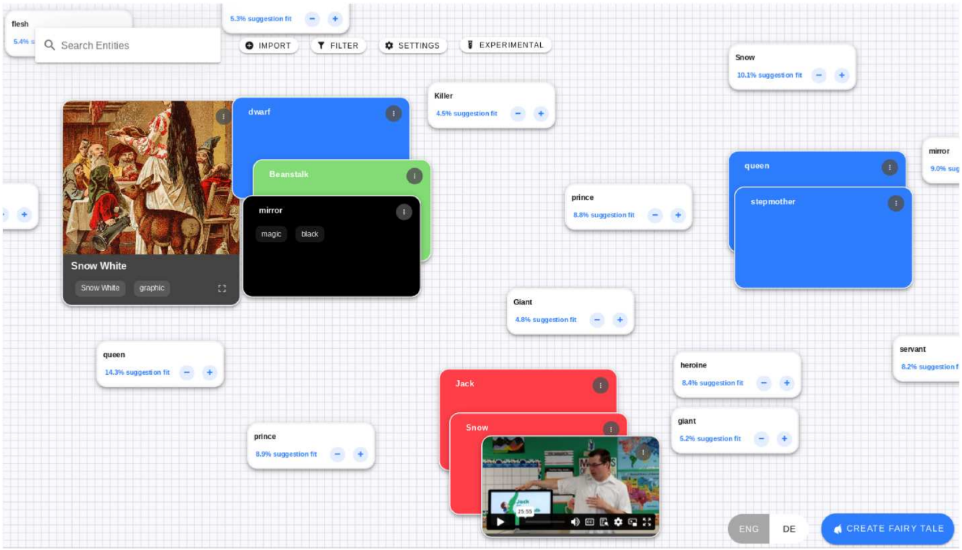


Figure 6.1: SPORE spatial interface. Visual grouping of narrative elements forms the structural basis for system suggestions [19].

Emergent-to-Formal Transition (P2)

SPORE partially realizes the emergent-to-formal transition. Early-stage story development ("storybreaking") occurs through informal clustering on the canvas. Meaning emerges through proximity and repeated rearrangement. However, stabilization remains primarily within the narrative task. While clusters guide prose generation, the system does not explicitly formalize stabilized groups into reusable structural representations beyond query extraction.

Thus, informal structure is preserved, but the transition toward persistent formal knowledge representations remains limited in scope.

Workspace-Mediated AI Guidance (P3)

SPORE strongly realizes workspace-mediated AI guidance. A key mechanism is the spatial parser, which translates visual arrangement into a weighted graph structure. This mechanism operationalizes Principle 3 by converting implicit spatial cues into machine-usable signals.

Figure 6.2 illustrates how seed articles are processed into entities and co-occurrence-based weighted relationships. These relationships form the underlying knowledge graph. When users cluster elements in the interface (Figure 6.1), the spatial parser detects proximity-based groupings and generates structured queries accordingly.

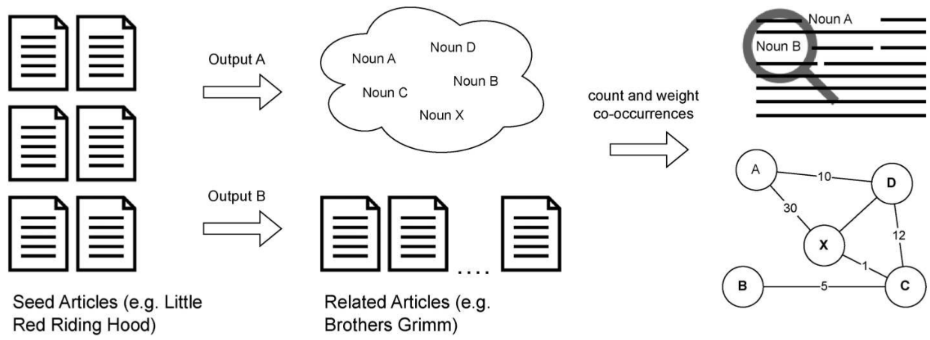


Figure 6.2: SPORE knowledge base construction: seed articles are processed into entity graphs through co-occurrence weighting [19].

Generative text production is then scoped by these spatially derived clusters. The prompt communicates which elements belong together, ensuring that prose generation reflects workspace structure rather than relying on unconstrained conversational context.

This explicit translation of spatial configuration into structured computational input represents one of the clearest realizations of mediation within the reviewed systems.

Iterative Co-Structuring (P4)

SPORE partially realizes iterative co-structuring. Authors repeatedly reorganize narrative elements, and suggestions adapt dynamically to evolving spatial context. However, generative output does not deeply restructure the workspace in return. The loop is interactive but primarily human-driven. The AI responds to spatial configuration rather than actively reshaping it.

Thus, iteration exists, but bidirectional structural evolution remains limited compared to the full cycle described in Chapter 5.

Mediation Layer

SPORE implements one of the most explicit mediation layers among the reviewed systems. The spatial parser converts visual layout into weighted graph representations, and structured queries are generated automatically from cluster detection. This mechanism bridges informal spatial hypertext and formal graph-based knowledge.

Unlike systems that rely solely on prompt encoding, SPORE introduces a computational extraction step that interprets spatial proximity quantitatively. This strengthens structural consistency and reduces reliance on textual prompt engineering.

Framework Interpretation

When evaluated against the hybrid intelligence framework, SPORE demonstrates strong alignment with human-centered control and workspace-mediated guidance. It provides concrete evidence that implicit spatial cues can be converted into machine-usable representations without eliminating user authority.

However, its domain focus on narrative co-creation limits generalizability. Formal stabilization beyond narrative scope remains underdeveloped, and iterative co-structuring is primarily reactive rather than fully bidirectional.

In summary, SPORE represents a structurally mature but domain-specific realization of hybrid intelligence. It strongly validates the mediation principle while only partially realizing long-term structural evolution across broader knowledge domains.

6.2.2 Case Analysis II: Sensecape

Sensecape [21] was developed to address a structural limitation of conventional LLM chat interfaces: the inability to maintain overview, revisit prior reasoning, and manage multiple abstraction levels during complex sensemaking tasks. Rather than integrating a formal knowledge graph, the system introduces a multilevel spatial workspace designed to externalize and stabilize evolving interpretation.

Human-Centered Structural Control (P1)

Sensecape strongly realizes human-centered structural control. Users create nodes, group them spatially, and define hierarchical relationships between canvases. The LLM produces suggestions or expansions when explicitly triggered, but it does not impose structural grouping.

Structure is shaped entirely through user interaction. This directly aligns with Principle 1 of the hybrid intelligence framework: generative output remains subordinate to human spatial organization.

Emergent-to-Formal Transition (P2)

Sensecape strongly realizes the emergent-to-formal transition. The system supports gradual movement from informal exploration to more stabilized abstraction layers.

Figure 6.3 illustrates this mechanism. In panel (a), content is presented at a flat semantic level ("all"), where detailed textual material dominates. In panel (b), semantic zoom reduces content to keyword-level representations, enabling users to identify conceptual anchors such as "Fisherman's Wharf" or "Clam Chowder." These elements can then be spatially linked and grouped across canvases.

This transition does not require predefined ontology. Instead, stabilization emerges through repeated grouping and hierarchical organization. Users progressively move from exploratory notes toward structured abstraction without being forced into rigid schemas. This directly reflects the emergent-to-formal transition defined in Chapter 5.

Workspace-Mediated AI Guidance (P3)

Sensecape partially realizes workspace-mediated AI guidance. AI operations are triggered from selected nodes or canvases, which localizes generative context. The spatial workspace therefore constrains generation indirectly.

However, unlike SPORE, Sensecape does not implement an explicit spatial parser that converts cluster configuration into structured graph representations. The AI responds to user selection but does not interpret spatial proximity or hierarchical structure as formal constraints. The workspace primarily supports human cognition rather than serving as a deeply computational substrate.

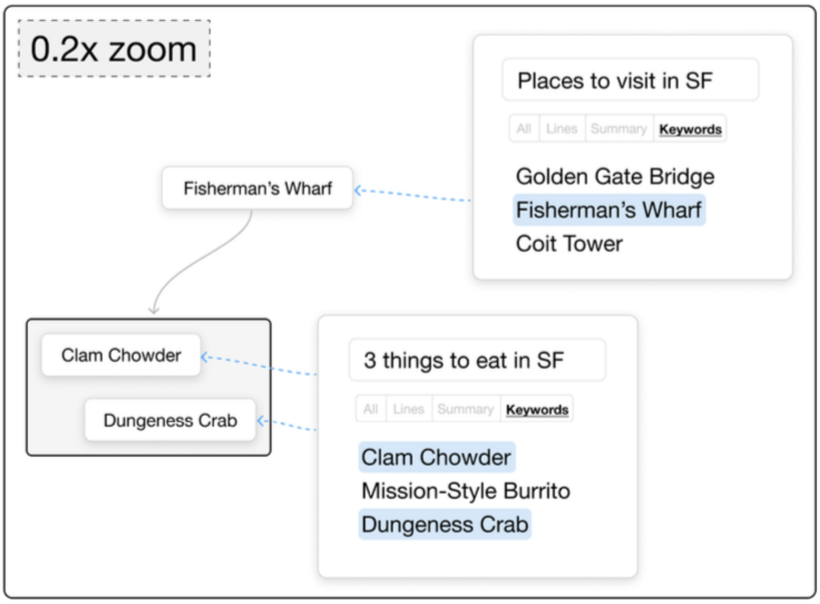
For this reason, workspace mediation is present but limited in structural depth.

Iterative Co-Structuring (P4)

Sensecape strongly realizes iterative co-structuring. The system is explicitly designed to support cycles of exploration, regrouping, abstraction, and refinement. Users can expand nodes, reorganize clusters, dive into sub-canvases, and revisit prior abstraction levels.

The empirical evaluation reported in [21] demonstrates increased breadth of exploration and deeper hierarchical representations compared to baseline chat-based systems. This supports the claim that spatial persistence enables sustained structural evolution.

(a) Semantic level: all



(b) Semantic level: keywords

Figure 6.3: Sensecape semantic zoom across abstraction levels. Panel (a) shows detailed content; panel (b) shows keyword-level abstraction and spatial linking [21].

However, the iteration remains primarily human-driven. AI-generated output does not autonomously reorganize workspace structure. The loop is sustained but not fully bidirectional.

Mediation Layer

Sensecape implements a lightweight mediation mechanism through node selection and abstraction-level triggering. Nevertheless, there is no explicit computational extraction of spatial relationships into weighted graphs or structured semantic representations.

Thus, while spatial organization strongly supports human interpretation and stabilization, its translation into machine-interpretable structure remains limited.

Framework Interpretation

Within the hybrid intelligence framework, Sensecape excels in supporting emergent stabilization and multilevel abstraction. It effectively addresses the structural limitations of linear chat interaction by introducing persistent, navigable spatial organization.

Its primary limitation lies in the absence of a robust mediation layer that transforms spatial configuration into structured computational constraints. The AI operates locally within selected scopes rather than reasoning over the full spatial structure.

Sensecape therefore represents a mature realization of spatially supported human–AI collaboration in terms of structural persistence and iterative organization, while leaving deeper structural mediation to future extensions.

6.2.3 Case Analysis III: Space-Steered LLM Summarization

Tang et al. [22] propose a system in which a human-constructed visual workspace is used to steer large language model summarization. Unlike conventional prompt-based summarization, this approach introduces an intermediate structural layer between human reasoning and generative output.

Human-Centered Structural Control (P1)

The system strongly reflects human-centered structural control. Before invoking the LLM, users construct a workspace containing filtered documents, clusters, highlights, annotations, and connections. The LLM does not determine grouping; it operates over structure that the human analyst has already externalized.

Figure 6.4 illustrates this mediation model. Documents and background expectations are first organized within a human workspace. The LLM receives not only documents but also structured signals derived from the workspace. The final report therefore depends on human-defined structural configuration.

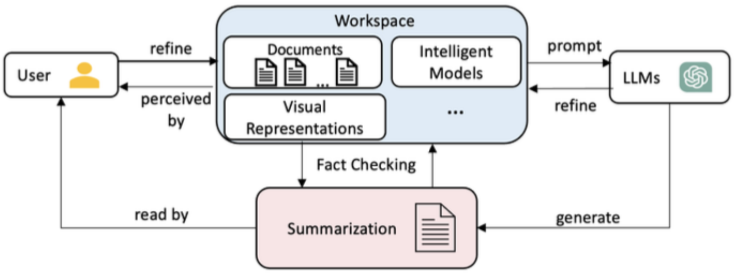


Figure 11: Workspace as a medium for human-AI collaboration for summarization.

Figure 6.4: Workspace as a medium for human–AI collaboration in summarization [22].

This architecture aligns directly with Principle 1 of the proposed hybrid intelligence framework: generative AI extends human structuring decisions but does not replace them.

Workspace-Mediated AI Guidance (P3)

[22] provides one of the clearest operational realizations of workspace-mediated AI guidance. Spatial clustering, filtering, highlighting, and annotation are translated into structured prompt representations. The workspace is not merely a visual aid; it becomes computational input.

Figure 6.5 further clarifies this pipeline. The workspace integrates documents, visual representations, and intelligent models. LLM generation is embedded within a feedback loop that includes refinement and fact checking. The user perceives and refines results through the workspace, making the workspace a shared representational layer.

Unlike chat-based systems where control depends on prompt wording alone, [22] reduces prompt fragility by embedding structural signals derived

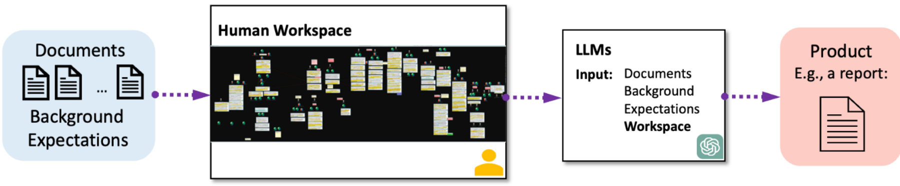


Figure 6.5: Workspace-centered architecture for human–AI summarization [22].

from spatial organization. This directly addresses the variability of probabilistic generation discussed in Chapter 2 [9].

Empirical Validation of Structural Mediation

A key contribution of [22] is quantitative evaluation. Figure 6.6 presents correctness scores across experimental conditions. Conditions that include filtering and clustering (E3, E7–E11) significantly outperform baseline document-only summarization (E1).

Prompts					Results					
ID	Documents	Filtering	Clustering	More Representations	Total Correctness Score	Average	Sub-item Scores: Who	When	Where	What
E1	√	-	-	-		34.4%	20%	27%	33%	56%
E2	√	√	-	-		45.5%	38%	19%	47%	67%
E3	√	√	√	-		65.2%	60%	77%	53%	73%
E4	√	-	-	Highlights		46.4%	51%	27%	27%	62%
E5	√	-	-	Annotates		48.8%	35%	92%	32%	54%
E6	√	-	-	Connections		44.4%	28%	69%	28%	61%
E7	√	√	√	Highlights		68.9%	75%	70%	47%	75%
E8	√	√	√	Annotates		69.1%	70%	88%	47%	72%
E9	√	√	√	Connections		62.6%	60%	72%	52%	68%
E10	√	√	√	Cluster Labels		75.9%	72%	19%	100%	95%
E11	√	√	√	All Above Except Connections		84.2%	92%	24%	100%	96%

Figure 6.6: Summarization performance across workspace configurations [22].

Notably, E11 (which combines filtering, clustering, and structural representations) achieves the highest average correctness score (84.2%). This empirically supports the theoretical claim developed in Chapters 3–5: externalized structure stabilizes generative output.

The results demonstrate that spatial mediation is not only conceptually meaningful but measurably improves generative alignment.

Emergent-to-Formal Transition (P2)

The system partially supports emergent-to-formal transition. Users begin by informally clustering and annotating documents. These spatial groupings are then translated into structured prompt inputs. However, [22] does not explicitly model long-term stabilization of clusters across extended iterative cycles. The workspace is optimized for summarization quality rather than progressive structural evolution.

Thus, P2 is present but task-bounded.

Iterative Co-Structuring (P4)

[22] supports limited iterative co-structuring. The architecture includes refinement loops between workspace and LLM (Figure 6.5), allowing users to adjust structure and regenerate summaries. However, iteration primarily improves summarization rather than reshaping the underlying conceptual structure over time.

Compared to Sensecape, which supports multilevel abstraction and exploration, [22]'s iteration is narrower and goal-oriented.

Framework Interpretation

Within the hybrid intelligence framework, [22] represents the strongest empirical validation of Principle 3 (workspace-mediated AI guidance). It clearly demonstrates that human-defined spatial structure improves LLM performance and reduces instability.

However, it does not fully realize iterative co-structuring as a sustained knowledge development process. The workspace mediates summarization tasks but does not function as a long-term evolving knowledge substrate.

[22] therefore exemplifies high structural mediation strength combined with moderate iterative depth. It confirms the theoretical claim that spatial organization can act as a stabilizing and guiding layer for generative AI, providing empirical grounding for the hybrid intelligence model proposed in Chapter 5.

6.2.4 Case Analysis IV: EDGE – Conversational Access to Structured Knowledge Graphs

EDGE (EDucational Knowledge Graph Explorer) [23] represents a different trajectory of integrating large language models into knowledge-based systems. Unlike spatial hypertext systems, EDGE combines conversational interfaces with formally structured knowledge graphs. The system enables users to query educational knowledge graphs through natural language, translating questions into Cypher queries executed on Neo4j databases.

Human-Centered Structural Control (P1)

EDGE partially realizes human-centered structural control. The user initiates queries in natural language and interprets returned results. However, structural organization is not constructed interactively during use. The ontology, entities, and relations are predefined in the knowledge graph schema.

This means that structural authority is distributed differently compared to spatial hypertext systems. In Sensecape or [22], users construct and refine structure through interaction. In EDGE, structure already exists in formalized graph form. The user navigates predefined semantics rather than shaping emergent conceptual clusters.

Therefore, P1 is realized in terms of interpretive authority, but not in terms of structural construction.

Emergent-to-Formal Transition (P2)

EDGE does not strongly realize the emergent-to-formal transition. The system operates over explicitly defined ontologies. Entities and relations are formalized prior to user interaction.

This contrasts sharply with spatial hypertext research [14, 17], where ambiguity and informal grouping support early-stage sensemaking. In EDGE, ambiguity is reduced through schema constraints. Knowledge stabilization occurs at design time, not through iterative user interaction.

From the hybrid intelligence perspective, EDGE represents the formal end of the structural spectrum. It supports reliable retrieval and deterministic response generation (temperature set to 0.0 in the reported implementation [23]), but it does not allow gradual structural emergence during use.

Workspace-Mediated AI Guidance (P3)

EDGE does not include a spatial workspace as a mediating layer. Instead, mediation occurs through schema-aware prompt construction. The LLM translates natural language into formal graph queries. The ontology constrains generation implicitly.

This represents a different form of structural mediation: ontology-driven rather than workspace-driven. While effective for precise retrieval, it lacks the flexible structural scaffolding characteristic of spatial hypertext environments.

In terms of the proposed framework, mediation exists, but it is rigid and predefined rather than emergent and visually externalized.

Iterative Co-Structuring (P4)

EDGE follows a traditional request–response paradigm. The interaction loop consists of:

1. User question
2. LLM translation into Cypher
3. Graph query execution
4. Natural language response

There is no persistent spatial representation where results can be clustered, reorganized, or reinterpreted visually. Iteration occurs conversationally but does not involve structural reconfiguration of a shared workspace.

As a result, iterative co-structuring is minimal. The system supports iterative querying, but not iterative structural stabilization.

Framework Interpretation

Within the hybrid intelligence framework, EDGE represents a strongly formalized integration model. It effectively combines LLM flexibility with explicit graph-based structure. This supports high interpretability and reliability for structured data retrieval.

However, it does not address the structural gap identified in Chapter 2: the lack of persistent, manipulable conceptual structure during exploratory knowledge work. Because structure is predefined, it cannot emerge through interaction.

EDGE therefore occupies the ontology-driven pole of hybrid intelligence systems. It demonstrates that generative AI can serve as a semantic interface to structured knowledge graphs. At the same time, it highlights what is missing without a spatial hypertext layer: tolerance for ambiguity, gradual stabilization, and visible structural co-construction.

This contrast is essential for the argument of this thesis. It shows that neither purely generative systems nor purely ontology-driven systems fully realize the hybrid intelligence model proposed in Chapter 5.

6.2.5 Case Analysis V: AISSH – An Exploratory AI-Supported Spatial Hypertext Prototype

AISSH (AI-Supported Spatial Hypertext) represents an exploratory prototype developed prior to the present thesis in order to investigate the technical feasibility of integrating generative AI with a spatial hypertext environment [?]. The system was implemented as a modular web application combining a browser-based spatial workspace with a generative chat interface.

Unlike SPORE or [22], AISSH was not designed around a specific domain task such as narrative construction or summarization. Instead, its purpose was architectural exploration: to examine how bidirectional interaction between a spatial hypertext canvas and a generative AI module could be realized in practice.

Human-Centered Structural Control (P1)

AISSH strongly reflects human-centered structural control. The spatial workspace allows users to create, move, cluster, and interpret nodes manually. Nodes can represent extracted entities, concepts from conversation, or manually inserted elements. Spatial grouping is entirely user-driven.

The generative AI component supports drafting, explanation, and entity extraction, but it does not autonomously restructure the workspace. Structural meaning remains under human control, consistent with Principle 1 of the hybrid intelligence framework.

Emergent-to-Formal Transition (P2)

AISSH partially supports emergent structure. Clusters form visually through spatial proximity and manual grouping. During early exploration, nodes remain loosely organized. Meaning stabilizes only through repeated repositioning and interpretation.

However, unlike SPORE, AISSH does not yet include a fully implemented spatial parser that translates visual grouping into weighted graph structures. Although relationship extraction modules were prepared and data structures were designed for Neo4j synchronization, full automated mediation between visual space and persistent graph structure was not completed.

This reveals a key architectural challenge: transforming informal spatial arrangement into computationally usable structural representations without forcing premature formalization.

Workspace-Mediated AI Guidance (P3)

AISSH includes a basic mediation mechanism. Before generating responses, the AI module receives serialized information about the current workspace state. This allows generation to incorporate contextual information beyond the immediate chat history.

However, mediation remains limited. Spatial proximity is not automatically interpreted as weighted relational structure. Clusters are recognized visually but not algorithmically translated into structured constraints. Compared to [22] or SPORE, workspace-mediated guidance is therefore present but shallow.

This limitation highlights the importance of the mediation layer defined in Chapter 5.

Iterative Co-Structuring (P4)

AISSH supports iterative interaction between chat and workspace. AI-generated responses can be transformed into new nodes. Extracted entities can be added to the canvas. Users can reorganize nodes and continue interaction based on updated spatial configuration.

This creates a bidirectional interaction pattern: conversation influences workspace, and workspace context influences conversation. While the cycle is not yet fully automated or optimized, it demonstrates practical feasibility of iterative co-structuring.

Importantly, the system treats AI output as provisional. Generated content becomes manipulable spatial material rather than final knowledge.

Framework Interpretation

Within the hybrid intelligence framework, AISSH can be interpreted as an early-stage architectural realization of the proposed principles. It demonstrates:

- Persistent spatial workspace as shared representational layer
- Human-centered structural authority
- Basic workspace-to-AI context mediation
- Iterative interaction between generation and spatial organization

At the same time, AISSH reveals open technical challenges:

- Automated extraction of implicit spatial relations

- Robust synchronization between visual clusters and graph databases
- Formal evaluation of structural stabilization effects

Unlike published systems optimized for specific tasks, AISSH exposes architectural tensions directly. It shows that hybrid intelligence integration is technically feasible but requires careful design of mediation layers and structural persistence mechanisms.

In this sense, AISSH serves as a transitional case between conceptual framework and implementation practice. It confirms that the theoretical model proposed in Chapter 5 is not merely abstract, but also technically realizable, while simultaneously illustrating the complexity of achieving deep structural integration.

6.3 Comparative Synthesis Through the Hybrid Intelligence Framework

The preceding case analyses show that elements of hybrid intelligence are already appearing across different systems. However, none of the examined systems fully realizes all four principles introduced in Chapter 5. To clarify their relative strengths and limits, Table 6.2 compares how each system aligns with the proposed framework.

Rather than presenting a simple classification, the comparison reveals a structural spectrum. At one end, EDGE represents ontology-driven integration, where structure is predefined, explicit, and machine-interpretable, but emergent stabilization and spatial co-structuring are largely absent. The system focuses on reliability and deterministic retrieval instead of exploratory flexibility. At the other end, Sensecape strongly supports emergent organization and multilevel abstraction. It addresses limitations of linear chat interaction by introducing persistent spatial structure, yet its mediation layer remains relatively lightweight, meaning that spatial configuration mainly supports human cognition rather than deep computational structuring. SPORE occupies a middle position by operationalizing spatial parsing and translating visual grouping into weighted graph representations. This makes it one of the most structurally advanced examples of workspace-mediated AI guidance, although its narrative co-creation focus limits broader generalization. [22] provides strong empirical evidence for structural mediation, showing through quantitative results that filtering, clustering, and structured workspace signals can significantly improve generative alignment. At the same time, its iterative process remains task-oriented rather than directed toward long-term

Table 6.2: Alignment of illustrative systems with the four hybrid intelligence principles.

System	P1: Human Structural Control	P2: Emergent-to-Formal Transition	P3: Workspace-Mediated Guidance	P4: Iterative Co-Structuring
SPORE [19]	Strong (user organizes story elements)	Moderate (clusters guide generation but domain-specific)	Strong (spatial parser converts layout into graph signals)	Moderate (limited long-term structural evolution)
Sensecape [21]	Strong (user defines hierarchy and grouping)	Strong (multilevel abstraction and semantic zoom)	Moderate (local node-based guidance)	Strong (iterative exploration and refinement)
[22]	Strong (workspace defines summarization scope)	Moderate (task-bound structural stabilization)	Strong (workspace translated into structured prompts)	Moderate (iteration focused on summarization quality)
EDGE [23]	Moderate (interpretive control but predefined schema)	Weak (structure formalized a priori)	Ontology-driven (schema-based mediation)	Weak (linear request–response loop)
AISSH [?]	Strong (fully user-controlled spatial structure)	Moderate (informal clusters, limited formalization)	Emerging (basic workspace-to-AI serialization)	Moderate (bidirectional but not fully automated)

structural evolution. AISSH demonstrates architectural feasibility by showing that bidirectional interaction between spatial hypertext and generative AI can be implemented within a modular system, while also revealing open challenges related to automated spatial parsing and synchronization between visual clusters and persistent graph representations.

Across all systems, a consistent pattern becomes visible: generative flexibility and structural persistence are rarely fully integrated. Systems tend to prioritize either formal structure without emergence or emergent organization without deep computational mediation. This observation directly supports the main argument of this thesis, namely that hybrid intelligence in knowledge-based systems requires a combination of persistent external structure, human-centered structural authority, a mediation layer that translates spatial configuration into computational context, and iterative co-structuring as the primary interaction mode. None of the reviewed systems fulfills all of these conditions at full strength. For this reason, the framework proposed in Chapter 5 should be understood not as a replacement for existing approaches but as a synthesis that brings together complementary insights currently distributed across different implementations.

This chapter examined representative systems using the analytical perspective introduced in Chapters 4 and 5. The analysis shows that spatial workspaces can help stabilize interaction with generative AI, that structured mediation can improve alignment, and that ontology-driven systems can increase reliability, although often at the cost of exploratory flexibility. At the same time, the reviewed systems do not realize these qualities in the same way, and each approach reflects different structural trade-offs. The next chapter brings together the theoretical and analytical findings of the thesis and discusses what they mean for hybrid intelligence in knowledge work. It also reflects on the limitations of the proposed framework and outlines possible directions for future research.

Chapter 7

Conclusion

7.1 Synthesis of Findings: Toward Structured Human–AI Collaboration

This thesis focused on building a theoretical framework that explains how generative AI and spatial hypertext can be used together to support human–AI collaboration in knowledge-based systems. The main issue discussed in the thesis is that generative AI, even though it has strong abilities, does not by itself create stable and interpretable knowledge structures.

Large Language Models can generate fluent text, summarize documents, and support drafting in many domains [3]. However, their outputs are probabilistic, they depend a lot on prompts, and they are not grounded in explicit, inspectable representations [7, 9]. They can expand and synthesize content, but they do not naturally turn knowledge into persistent structures that users can easily review, change, and refine.

At the same time, research on spatial hypertext shows that knowledge work is usually not linear and not fully planned from the start. It develops through grouping, rearranging, and gradual stabilization of meaning [14, 15]. Spatial hypertext replaces fixed link structures with spatial and visual cues that helps users show fictional relationships and reorganize them over the time [13, 14]. This kind of flexible work supports ambiguity and changing interpretation, which are important during early-stage sensemaking [17].

When these two research directions are considered together, a structural asymmetry becomes clear. Generative AI is strong at producing content but it lacks stable structure. Spatial hypertext supports emergent, human-controlled structure, but it does not generate content on its own. Recent systems such as Sensecape and visual steering approaches show that spatial organization can affect AI outputs [21, 22]. Research on hybrid intelligence also shows that effective collaboration depends on a complementary division of cognitive labor rather than full automation [8].

Based on this background, the thesis addressed three research questions.

RQ1: How can generative AI and spatial hypertext be structurally combined?

The results show that integration happens through an iterative co-structuring process. In this process, AI produces candidate content, the human arranges and interprets it spatially, and the resulting structure becomes contextual guidance for the next generation step. Collaboration develops through repeated cycles of generation, arrangement, and refinement instead of single prompt–response interactions.

RQ2: What roles do spatial organization and emergent structure play?

The analysis shows that spatial organization is not just a visual interface element. It works as a key mechanism for stabilizing developing knowledge. Emergent clustering allows users to handle ambiguity and delay strict formalization [14, 17]. In hybrid systems, this emergent structure becomes a persistent scaffold for AI-generated content, supporting a gradual shift from informal grouping to more explicit representations.

RQ3: How can spatial hypertext function as a shared cognitive workspace?

The workspace acts as a shared representational layer. For the human, it externalizes reasoning, priorities, and interpretations [15]. For the AI, it can act as structured contextual input that limits scope and guides emphasis [22]. In this way, the workspace supports interaction, keeps structural anchors across iterations, and makes the collaborative process easier to follow.

Overall, the combined findings from Chapters 2 through 6 lead to one main conclusion: generative AI increases the ability to produce and synthesize knowledge, while spatial hypertext stabilizes and structures that knowledge through human-centered organization. Effective human–AI collaboration does not come from replacing human judgment with automatic generation, but from coordinating generative flexibility with stable and interpretable spatial structure.

7.2 Theoretical Contributions and Practical Implications

This thesis mainly contributes on a theoretical and structural level. It does not introduce new algorithms or performance benchmarks. Instead, it explains in a clearer way how generative AI and spatial hypertext can be integrated in a principled way.

The first contribution is that the thesis identifies and explains a structural asymmetry between generative AI and spatial hypertext. Generative AI provides scale, speed, and synthesis, but it still does not provide persistent and inspectable structure [7, 9]. Spatial hypertext supports emergent, human-controlled structure, but it does not have generative capacity [13, 14]. By explaining this complementarity more directly, the thesis positions spatial hypertext as a mediation layer between human cognition and machine generation. The second contribution is the definition of four integration principles: human-centered structural control, emergent-to-formal transition, workspace-mediated AI guidance, and iterative co-structuring. These principles are not only general claims about "AI collaboration". Instead, they describe more concrete structural conditions for hybrid systems. They explain how responsibilities are shared between the human and the AI, how structure develops over time, and how interaction cycles should be organized. The third contribution is the conceptual architecture that applies these principles. By separating the human agent, generative module, spatial workspace, and mediation layer, the framework clarifies how hybrid systems can keep interpretability while still using generative power. In addition to the theory, the thesis also points to practical implications. The analysis of existing systems shows that workspace-derived structural signals can improve generative alignment [22]. Multilevel spatial exploration can increase breadth and revisitation in complex tasks [21]. Graph-based systems such as EDGE show the benefits of explicit structure, but they also show the limits of predefined ontologies when work is exploratory. Based on these observations, some implications for system designers follow. Relying only on prompts is not sufficient for complex knowledge tasks. Persistent spatial workspaces can stabilize developing reasoning. Structural authority should stay with the human user. Formalization should happen gradually and not too early. Overall, these points show that integrating generative AI with spatial hypertext is not only theoretically but also practically useful for knowledge-based systems.

7.3 Limitations and Future Research Directions

To conclude, several limitations should be mentioned. Firstly, the proposed framework is abstract. It is grounded in literature and comparative system analysis, but it has not been validated through controlled experimental implementation. Because of this, the framework should be understood as a theoretical model and not as an empirically tested architecture. Secondly, spatial mediation mechanisms are still under-specified at the algorithmic level. Systems such as SPORE and [22] these provide examples of spatial parsing and structured prompt translation [19, 22], but a general mechanism that can interpret implicit spatial cues across different domains is still an open challenge. Third, scalability creates design tensions. Spatial hypertext research shows that purely spatial organization can become difficult to manage when complexity increases [17]. Multilevel approaches such as Sensecape reduce this issue [21], but long-term knowledge bases may still require additional structuring mechanisms. Fourth, usability trade-offs also need to be considered. Large spatial interaction and multilevel abstractions can increase the learning effort for users. Due to this, hybrid systems need to balance with cognitive clarity and expressive power.

Future research can tackle these limitations in different ways. One way is to develop flexible spatial interpretation mechanisms that preserve informality while still enabling computational use of clustering, hierarchy, and selection signals. Another way is reliability and grounding, for example by adding fact-checking layers and better source transparency to reduce hallucination risks in generative models [7, 9]. Long-term studies are also needed to understand how hybrid workspaces support knowledge development over longer time periods. Additionally, collaborative multi-user environments are an important extension, where shared spatial structures can evolve through interaction between multiple humans and AI systems. Finally, future systems could inspect gradual formalization layers inspired by architectures such as Mother [18]. Informal clusters could slowly transition into more direct relational structures once stability and shared understanding emerge.

Generative AI and spatial hypertext focus on different but complementary aspects of knowledge work. Generative AI expands knowledge through fast synthesis, while spatial hypertext stabilizes knowledge through visible and persistent structure. When integrated through carefully designed hybrid approaches, these paradigms can support structured human–AI collaboration based on interpretability, flexibility, and shared control.

List of Figures

List of Tables

List of Abbreviations

AI	Artificial Intelligence
GAI	Generative Artificial Intelligence
LLM	Large Language Model
NLP	Natural Language Processing
NLU	Natural Language Understanding
NLG	Natural Language Generation
DM	Dialogue Management
GAE	Generative Artificial Expert
VAE	Variational Autoencoder
GAN	Generative Adversarial Network
API	Application Programming Interface
KG	Knowledge Graph
SH	Spatial Hypertext

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